GRAVITATIONAL FIELDS

13.1 Gravitational field

Candidates should be able to:

1 understand that a gravitational field is an example of a field of force and define gravitational field as force per unit mass

2 represent a gravitational field by means of field lines

13.2 Gravitational force between point masses

Candidates should be able to:

1 understand that, for a point outside a uniform sphere, the mass of the sphere may be considered to be a point mass at its centre

2 recall and use Newton's law of gravitation $F = Gm_1m_2/r^2$ for the force between two point masses

3 analyse circular orbits in gravitational fields by relating the gravitational force to the centripetal acceleration it causes

4 understand that a satellite in a geostationary orbit remains at the same point above the Earth's surface, with an orbital period of 24 hours, orbiting from west to east, directly above the Equator

13.3 Gravitational field of a point mass

Candidates should be able to:

1 derive, from Newton's law of gravitation and the definition of gravitational field, the equation $g = GM/r^2$ for the gravitational field strength due to a point mass

2 recall and use $g = GM/r^2$

3 understand why g is approximately constant for small changes in height near the Earth's surface

13.4 Gravitational potential

Candidates should be able to:

1 define gravitational potential at a point as the work done per unit mass in bringing a small test mass from infinity to the point

2 use $\phi = -GM/r$ for the gravitational potential in the field due to a point mass

3 understand how the concept of gravitational potential leads to the gravitational potential energy of two point masses and use $E_p = -GMm/r$

1. M/J/07/Q1

(a) Explain what is meant by a *gravitational field*.

....

.... [1]

(b) A spherical planet has mass M and radius R. The planet may be considered to have all its mass concentrated at its centre.
A rocket is launched from the surface of the planet such that the rocket moves radially away from the planet. The rocket engines are stopped when the rocket is at a height R above the surface of the planet, as shown in Fig. 1.1.

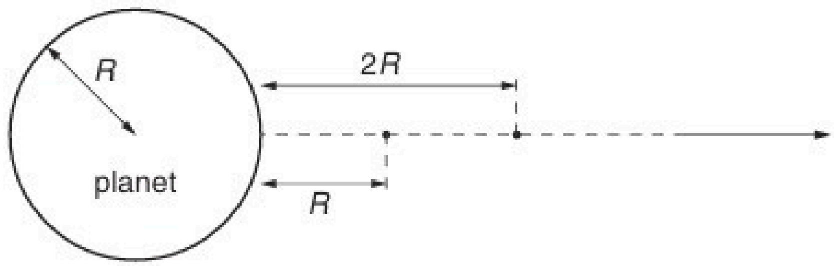


Fig. 1.1

The mass of the rocket, after its engines have been stopped, is m.

(i) Show that, for the rocket to travel from a height R to a height $2R$ above the planet's surface, the change ΔE_P in the magnitude of the gravitational potential energy of the rocket is given by the expression

$$\Delta E_P = \frac{GMm}{6R}.$$

[2]

(ii) During the ascent from a height R to a height $2R$, the speed of the rocket changes from $7600\,\mathrm{m\,s^{-1}}$ to $7320\,\mathrm{m\,s^{-1}}$. Show that, in SI units, the change ΔE_K in the kinetic energy of the rocket is given by the expression

$$\Delta E_K = (2.09 \times 10^6)m.$$

[1]

(c) The planet has a radius of $3.40 \times 10^6\,\mathrm{m}$.

(i) Use the expressions in **(b)** to determine a value for the mass M of the planet.

M = ... kg [2]

(ii) State one assumption made in the determination in **(i)**.

....

.... [1]

2. M/J/09/Q1

(a) Define *gravitational field strength.*

....

.... [1]

(b) A spherical planet has diameter 1.2×10^4 km. The gravitational field strength at the surface of the planet is $8.6\,\mathrm{N\,kg^{-1}}$.
The planet may be assumed to be isolated in space and to have its mass concentrated at its centre.
Calculate the mass of the planet.

mass = ... kg [3]

(c) The gravitational potential at a point X above the surface of the planet in **(b)** is $-5.3 \times 10^7\,\mathrm{J\,kg^{-1}}$.
For point Y above the surface of the planet, the gravitational potential is $-6.8 \times 10^7\,\mathrm{J\,kg^{-1}}$.

(i) State, with a reason, whether point X or point Y is nearer to the planet.

....

....

.... [2]

(ii) A rock falls radially from rest towards the planet from one point to the other.
Calculate the final speed of the rock.

speed = ... $\mathrm{m\,s^{-1}}$ [2]

3. O/N/09/42/Q1

(a) The Earth may be considered to be a uniform sphere of radius 6.38×10^3 km, with its mass concentrated at its centre.

(i) Define *gravitational field strength*.

....

.... [1]

(ii) By considering the gravitational field strength at the surface of the Earth, show that the mass of the Earth is 5.99×10^{24} kg.

[2]

(b) The Global Positioning System (GPS) is a navigation system that can be used anywhere on Earth. It uses a number of satellites that orbit the Earth in circular orbits at a distance of 2.22×10^4 km above its surface.

(i) Use data from **(a)** to calculate the angular speed of a GPS satellite in its orbit.

angular speed = ... $rad\,s^{-1}$ [3]

(ii) Use your answer in **(i)** to show that the satellites are not in geostationary orbits.

[3]

(c) The planes of the orbits of the GPS satellites in **(b)** are inclined at an angle of 55° to the Equator.

Suggest why the satellites are not in equatorial orbits.

....

.... [1]

4. O/N/09/41/Q1

(a) State Newton's law of gravitation.

....

....

.... [2]

(b) The Earth may be considered to be a uniform sphere of radius R equal to 6.4×10^6 m.

A satellite is in a geostationary orbit.

(i) Describe what is meant by a *geostationary orbit*.

....

....

....

.... [3]

(ii) Show that the radius x of the geostationary orbit is given by the expression

$$gR^2 = x^3\omega^2$$

where g is the acceleration of free fall at the Earth's surface and ω is the angular speed of the satellite about the centre of the Earth.

[3]

(iii) Determine the radius x of the geostationary orbit.

radius = ... m [3]

5. M/J/10/42/Q1

(a) Define *gravitational potential* at a point.

....

....

.... [2]

(b) The Earth may be considered to be an isolated sphere of radius R with its mass concentrated at its centre.
The variation of the gravitational potential ϕ with distance x from the centre of the Earth is shown in Fig. 1.1.

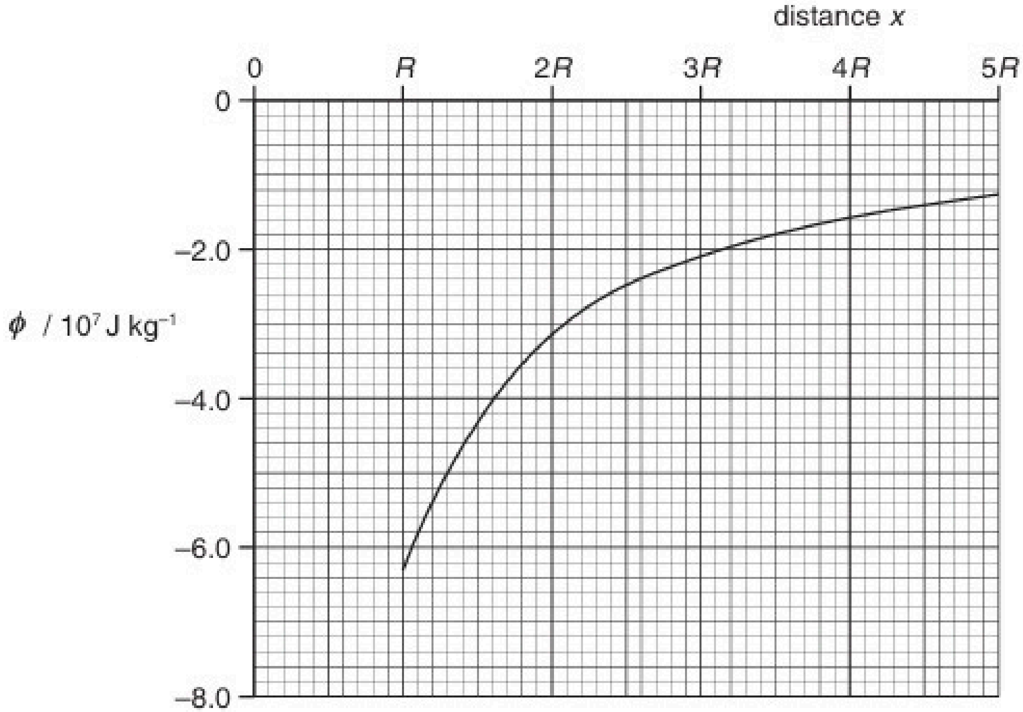


Fig. 1.1

The radius R of the Earth is 6.4×10^6 m.

(i) By considering the gravitational potential at the Earth's surface, determine a value for the mass of the Earth.

mass = ... kg [3]

(ii) A meteorite is at rest at infinity. The meteorite travels from infinity towards the Earth.

Calculate the speed of the meteorite when it is at a distance of $2R$ above the Earth's surface. Explain your working.

speed = ... $m\,s^{-1}$ [4]

(iii) In practice, the Earth is not an isolated sphere because it is orbited by the Moon, as illustrated in Fig. 1.2.

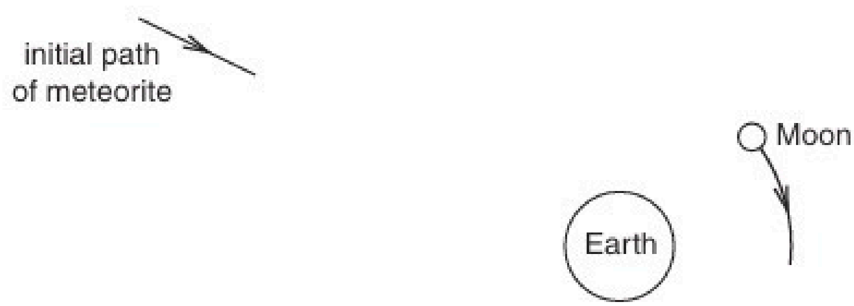


Fig. 1.2 (not to scale)

The initial path of the meteorite is also shown.

Suggest two changes to the motion of the meteorite caused by the Moon.

1. ....

....

2. ....

....

[2]

6. O/N/10/42/Q1

(a) Define *gravitational field strength*.

....

.... [1]

(b) An isolated star has radius R. The mass of the star may be considered to be a point mass at the centre of the star.
The gravitational field strength at the surface of the star is g_s.

On Fig. 1.1, sketch a graph to show the variation of the gravitational field strength of the star with distance from its centre. You should consider distances in the range R to $4R$.

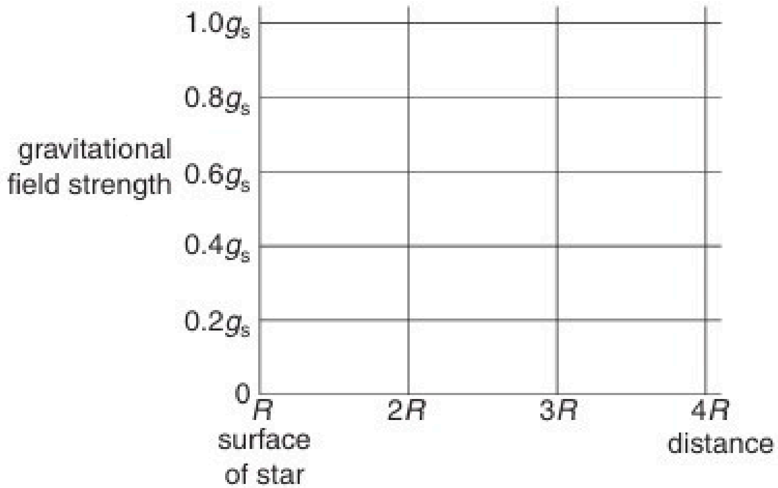


Fig. 1.1

[2]

(c) The Earth and the Moon may be considered to be spheres that are isolated in space with their masses concentrated at their centres.
The masses of the Earth and the Moon are 6.00×10^{24} kg and 7.40×10^{22} kg respectively.
The radius of the Earth is R_E and the separation of the centres of the Earth and the Moon is 60 R_E, as illustrated in Fig. 1.2.

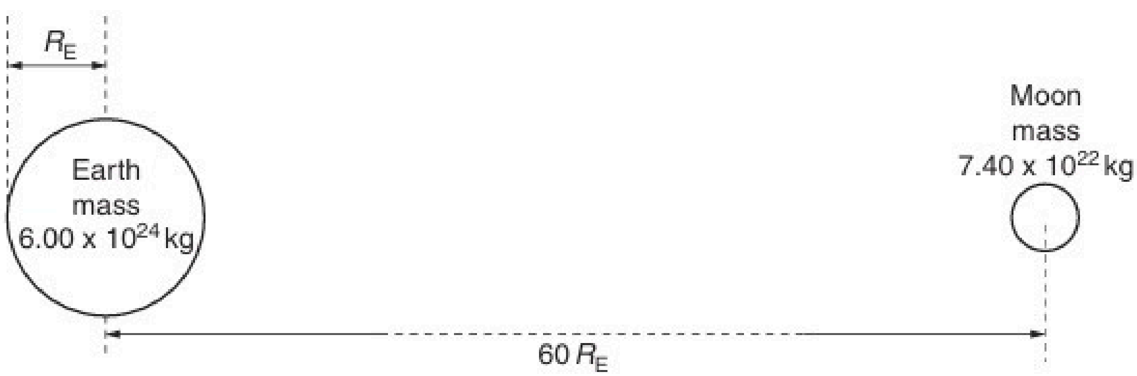


Fig. 1.2 (not to scale)

(i) Explain why there is a point between the Earth and the Moon at which the gravitational field strength is zero.

....

....

.... [2]

(ii) Determine the distance, in terms of R_E, from the centre of the Earth at which the gravitational field strength is zero.

distance = ... R_E [3]

(iii) On the axes of Fig. 1.3, sketch a graph to show the variation of the gravitational field strength with position between the surface of the Earth and the surface of the Moon.

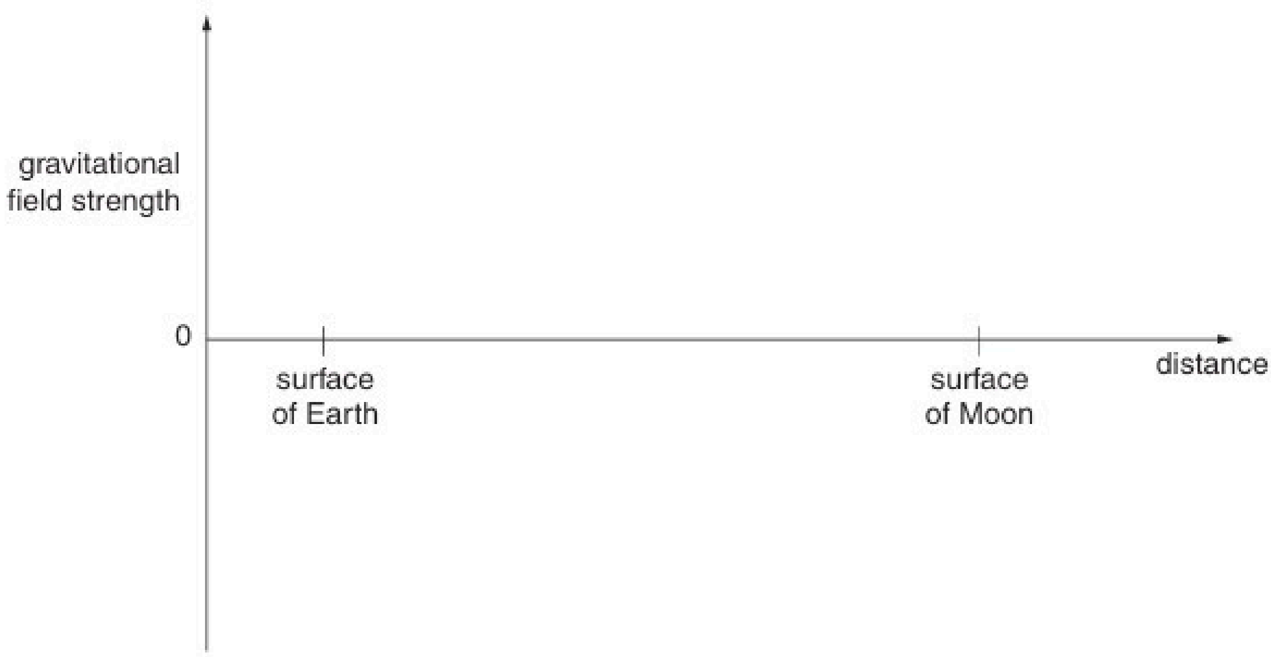


Fig. 1.3

[3]

7. M/J/11/41/Q1

(a) Newton's law of gravitation applies to point masses.

(i) State Newton's law of gravitation.

....

....

.... [2]

(ii) Explain why, although the planets and the Sun are not point masses, the law also applies to planets orbiting the Sun.

....

.... [1]

(b) Gravitational fields and electric fields show certain similarities and certain differences. State one aspect of gravitational and electric fields where there is

(i) a similarity,

....

.... [1]

(ii) a difference.

....

....

.... [2]

8. M/J/12/41/Q1

(a) Define *gravitational potential* at a point.

....

.... [1]

(b) The gravitational potential ϕ at distance r from point mass M is given by the expression

$$\phi = -\frac{GM}{r}$$

where G is the gravitational constant.

Explain the significance of the negative sign in this expression.

....

....

.... [2]

(c) A spherical planet may be assumed to be an isolated point mass with its mass concentrated at its centre. A small mass m is moving near to, and normal to, the surface of the planet. The mass moves away from the planet through a short distance h.

State and explain why the change in gravitational potential energy ΔE_P of the mass is given by the expression

$$\Delta E_P = mgh$$

where g is the acceleration of free fall.

....

....

....

....

....

.... [4]

(d) The planet in **(c)** has mass M and diameter 6.8×10^3 km. The product GM for this planet is $4.3 \times 10^{13}\,\mathrm{N\,m^2\,kg^{-1}}$.

A rock, initially at rest a long distance from the planet, accelerates towards the planet. Assuming that the planet has negligible atmosphere, calculate the speed of the rock as it hits the surface of the planet.

speed = ... $\mathrm{m\,s^{-1}}$ [3]

9. M/J/12/42/Q1

(a) State Newton's law of gravitation.

....

....

.... [2]

(b) The Earth and the Moon may be considered to be isolated in space with their masses concentrated at their centres.
The orbit of the Moon around the Earth is circular with a radius of 3.84×10^5 km. The period of the orbit is 27.3 days.

Show that

(i) the angular speed of the Moon in its orbit around the Earth is $2.66 \times 10^{-6}\,\text{rad}\,\text{s}^{-1}$,

[1]

(ii) the mass of the Earth is 6.0×10^{24} kg.

[2]

(c) The mass of the Moon is 7.4×10^{22} kg.

(i) Using data from **(b)**, determine the gravitational force between the Earth and the Moon.

force = ... N [2]

(ii) Tidal action on the Earth's surface causes the radius of the orbit of the Moon to increase by 4.0 cm each year.

Use your answer in **(i)** to determine the change, in one year, of the gravitational potential energy of the Moon. Explain your working.

energy change = ... J [3]

10. O/N/12/42/Q1

(a) State Newton's law of gravitation.

....

....

.... [2]

(b) A satellite of mass m is in a circular orbit of radius r about a planet of mass M.
For this planet, the product GM is $4.00 \times 10^{14}\,\mathrm{N\,m^2\,kg^{-1}}$, where G is the gravitational constant.
The planet may be assumed to be isolated in space.

(i) By considering the gravitational force on the satellite and the centripetal force, show that the kinetic energy E_K of the satellite is given by the expression

$$E_K = \frac{GMm}{2r}.$$

[2]

(ii) The satellite has mass 620 kg and is initially in a circular orbit of radius 7.34×10^6 m, as illustrated in Fig. 1.1.

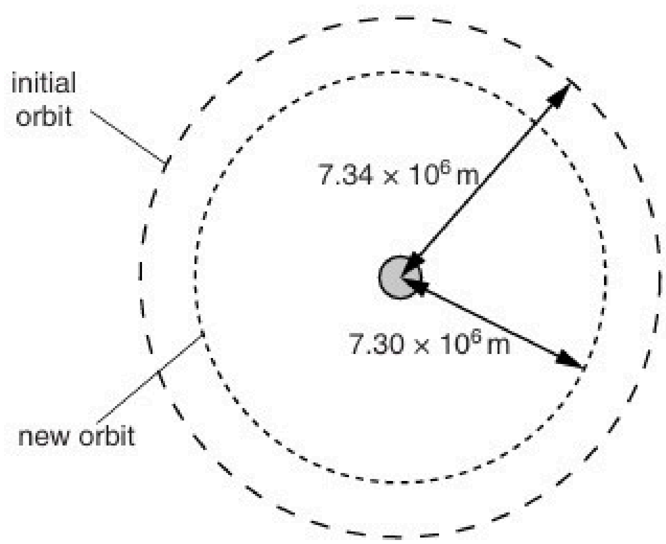


Fig. 1.1 (not to scale)

Resistive forces cause the satellite to move into a new orbit of radius 7.30×10^{6} m.

Determine, for the satellite, the change in

1. kinetic energy,

change in kinetic energy = ... J [2]

2. gravitational potential energy.

change in potential energy = ... J [2]

(iii) Use your answers in **(ii)** to explain whether the linear speed of the satellite increases, decreases or remains unchanged when the radius of the orbit decreases.

....

....

....[2]

11. M/J/13/41/Q1

(a) State what is meant by a *gravitational field*.

....

....

.... [2]

(b) In the Solar System, the planets may be assumed to be in circular orbits about the Sun. Data for the radii of the orbits of the Earth and Jupiter about the Sun are given in Fig. 1.1.

	radius of orbit / km
Earth	1.50×10^8
Jupiter	7.78×10^8

Fig. 1.1

(i) State Newton's law of gravitation.

....

....

....

.... [3]

(ii) Use Newton's law to determine the ratio

$$\frac{\text{gravitational field strength due to the Sun at orbit of Earth}}{\text{gravitational field strength due to the Sun at orbit of Jupiter}}.$$

ratio = ... [3]

(c) The orbital period of the Earth about the Sun is T.

(i) Use ideas about circular motion to show that the mass M of the Sun is given by

$$M = \frac{4\pi^2 R^3}{GT^2}$$

where R is the radius of the Earth's orbit about the Sun and G is the gravitational constant.
Explain your working.

[3]

(ii) The orbital period T of the Earth about the Sun is 3.16×10^7 s.
The radius of the Earth's orbit is given in Fig. 1.1.
Use the expression in **(i)** to determine the mass of the Sun.

mass = ... kg [2]

12. M/J/13/42/Q1

(a) Explain what is meant by a *geostationary orbit*.

....

....

....

.... [3]

(b) A satellite of mass m is in a circular orbit about a planet.
The mass M of the planet may be considered to be concentrated at its centre.
Show that the radius R of the orbit of the satellite is given by the expression

$$R^3 = \left(\frac{GMT^2}{4\pi^2}\right)$$

where T is the period of the orbit of the satellite and G is the gravitational constant.
Explain your working.

[4]

(c) The Earth has mass 6.0×10^{24} kg. Use the expression given in **(b)** to determine the radius of the geostationary orbit about the Earth.

radius = ... m [3]

13. O/N/13/41/Q1

(a) Define *gravitational potential* at a point.

....

....

.... [2]

(b) The Moon may be considered to be an isolated sphere of radius 1.74×10^3 km with its mass of 7.35×10^{22} kg concentrated at its centre.

(i) A rock of mass 4.50 kg is situated on the surface of the Moon. Show that the change in gravitational potential energy of the rock in moving it from the Moon's surface to infinity is 1.27×10^7 J.

[1]

(ii) The escape speed of the rock is the minimum speed that the rock must be given when it is on the Moon's surface so that it can escape to infinity.
Use the answer in **(i)** to determine the escape speed. Explain your working.

speed = ... $m s^{-1}$ [2]

(c) The Moon in **(b)** is assumed to be isolated in space. The Moon does, in fact, orbit the Earth.
State and explain whether the minimum speed for the rock to reach the Earth from the surface of the Moon is different from the escape speed calculated in **(b)**.

....

....

.... [2]

14. O/N/13/43/Q1

(a) State Newton's law of gravitation.

....

....

.... [2]

(b) A star and a planet are isolated in space. The planet orbits the star in a circular orbit of radius R, as illustrated in Fig. 1.1.

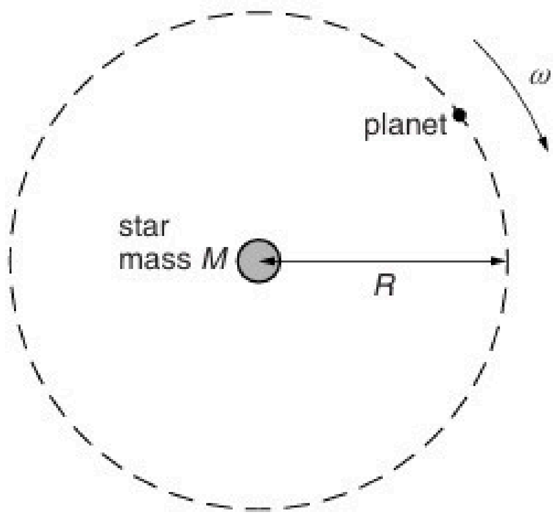


Fig. 1.1

The angular speed of the planet about the star is ω.
By considering the circular motion of the planet about the star of mass M, show that ω and R are related by the expression

$$R^3\omega^2 = GM$$

where G is the gravitational constant. Explain your working.

[3]

(c) The Earth orbits the Sun in a circular orbit of radius 1.5×10^8 km. The mass of the Sun is 2.0×10^{30} kg.
A distant star is found to have a planet that has a circular orbit about the star. The radius of the orbit is 6.0×10^8 km and the period of the orbit is 2.0 years.

Use the expression in **(b)** to calculate the mass of the star.

mass = ... kg [3]

15. M/J/14/41/Q1

(a) Define *gravitational potential* at a point.

....

....

.... [2]

(b) A stone of mass m has gravitational potential energy E_P at a point X in a gravitational field. The magnitude of the gravitational potential at X is ϕ.

State the relation between m, E_P and ϕ.

.... [1]

(c) An isolated spherical planet of radius R may be assumed to have all its mass concentrated at its centre. The gravitational potential at the surface of the planet is $-6.30 \times 10^7\,\text{J kg}^{-1}$.

A stone of mass 1.30 kg is travelling towards the planet such that its distance from the centre of the planet changes from $6R$ to $5R$.

Calculate the change in gravitational potential energy of the stone.

change in energy = ... J [4]

_16. M/J/14/42/Q1_

The mass M of a spherical planet may be assumed to be a point mass at the centre of the planet.

(a) A stone, travelling at speed v, is in a circular orbit of radius r about the planet, as illustrated in Fig.1.1.

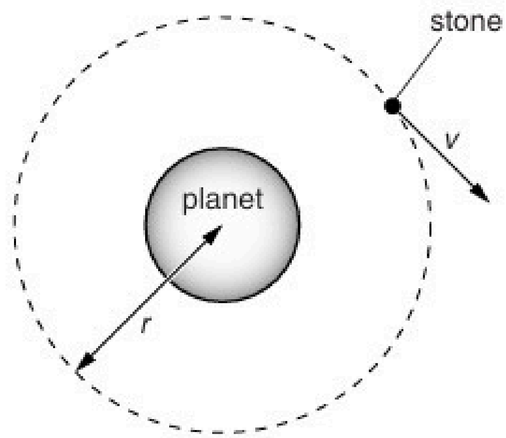


Fig.1.1

Show that the speed v is given by the expression

$$v = \sqrt{\left(\frac{GM}{r}\right)}$$

where G is the gravitational constant.
Explain your working.

[2]

(b) A second stone, initially at rest at infinity, travels towards the planet, as illustrated in Fig. 1.2.

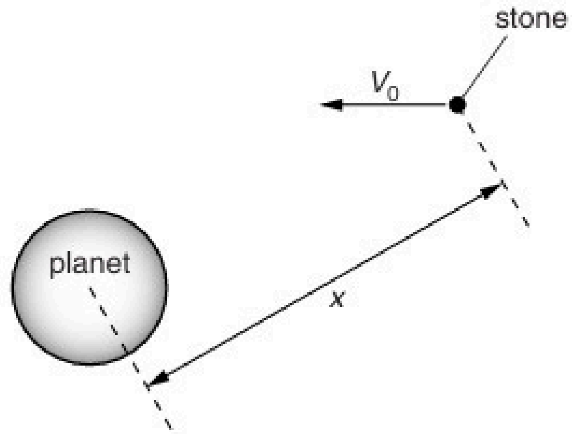


Fig. 1.2 (not to scale)

The stone does not hit the surface of the planet.

(i) Determine, in terms of the gravitational constant G and the mass M of the planet, the speed V_0 of the stone at a distance x from the centre of the planet. Explain your working. You may assume that the gravitational attraction on the stone is due only to the planet.

[3]

(ii) Use your answer in **(i)** and the expression in **(a)** to explain whether this stone could enter a circular orbit about the planet.

....

....

....

.... [2]

17. O/N/14/41/Q1

An isolated spherical planet has a diameter of 6.8×10^{6} m. Its mass of 6.4×10^{23} kg may be assumed to be a point mass at the centre of the planet.

(a) Show that the gravitational field strength at the surface of the planet is $3.7\,N\,kg^{-1}$.

[2]

(b) A stone of mass 2.4 kg is raised from the surface of the planet through a vertical height of 1800 m.
Use the value of field strength given in **(a)** to determine the change in gravitational potential energy of the stone.
Explain your working.

change in energy = ... J [3]

(c) A rock, initially at rest at infinity, moves towards the planet. At point P, its height above the surface of the planet is 3.5 D, where D is the diameter of the planet, as shown in Fig. 1.1.

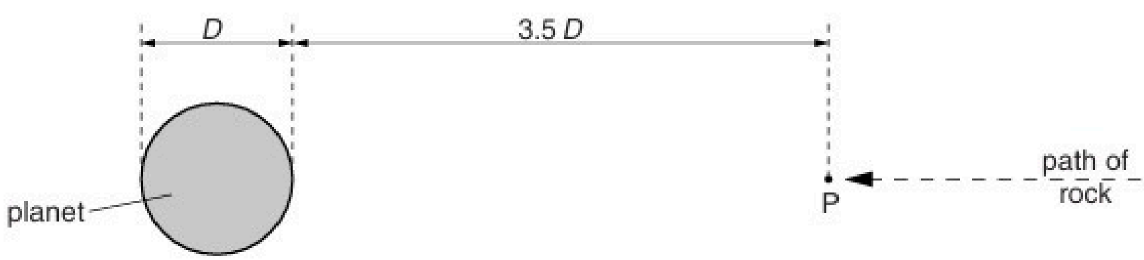


Fig. 1.1

Calculate the speed of the rock at point P, assuming that the change in gravitational potential energy is all transferred to kinetic energy.

speed = ... $m s^{-1}$ [4]

18. M/J/15/41/Q1

(a) State Newton's law of gravitation.

....

....

.... [2]

(b) The planet Neptune has eight moons (satellites). Each moon orbits Neptune in a circular path of radius r with a period T.

Assuming that Neptune and each moon behave as point masses, show that r and T are related by the expression

$$GM_N = \frac{4\pi^2 r^3}{T^2}$$

where G is the gravitational constant and M_N is the mass of Neptune.

[3]

(c) Data for the moon Triton that orbits Neptune and for the moon Oberon that orbits the planet Uranus are given in Fig. 1.1.

planet	moon	radius of orbit $r/10^5$ km	period of orbit T/days
Neptune	Triton	3.55	5.9
Uranus	Oberon	5.83	13.5

Fig. 1.1

Use the expression in **(b)** to determine the ratio

$$\frac{\text{mass of Neptune}}{\text{mass of Uranus}}.$$

ratio = ... [3]

19. M/J/15/42/Q1

(a) The Earth may be considered to be a uniform sphere of radius 6.37×10^3 km with its mass of 5.98×10^{24} kg concentrated at its centre. The Earth spins on its axis with a period of 24.0 hours.

(i) A stone of mass 2.50 kg rests on the Earth's surface at the Equator.

1. Calculate, using Newton's law of gravitation, the gravitational force on the stone.

gravitational force = ... N [2]

2. Determine the force required to maintain the stone in its circular path.

force = ... N [2]

(ii) The stone is now hung from a newton-meter.

Use your answers in **(i)** to determine the reading on the meter. Give your answer to three significant figures.

reading = ... N [2]

(b) A satellite is orbiting the Earth. For an astronaut in the satellite, his sensation of weight is caused by the contact force from his surroundings.

The astronaut reports that he is ‘weightless’, despite being in the Earth’s gravitational field.

Suggest what is meant by the astronaut reporting that he is ‘weightless’.

....

....

....

....[3]

20. O/N/15/41/Q1

A satellite of mass m_S is in a circular orbit of radius x about the Earth.

The Earth may be considered to be an isolated uniform sphere with its mass M concentrated at its centre.

(a) **(i)** Show that the kinetic energy E_K of the satellite is given by the expression

$$E_K = \frac{GMm_S}{2x}$$

where G is the gravitational constant. Explain your working.

[3]

(ii) State an expression, in terms of G, M, m_S and x, for the potential energy E_P of the satellite.

....[1]

(iii) Using answers from **(i)** and **(ii)**, derive an expression for the total energy E_T of the satellite.

E_T = ...[2]

(b) Small resistive forces acting on the satellite cause the radius of its circular orbit to change.

Use your answers in **(a)** to state, for the satellite, whether each of the following quantities increases, decreases or remains constant.

(i) total energy

....[1]

(ii) radius of orbit

....[1]

(iii) potential energy

....[1]

(iv) kinetic energy

....[1]

21. O/N/15/43/Q1

(a) State Newton's law of gravitation.

....

....

.... [2]

(b) Some of the planets in the Solar System have several moons (satellites) that have circular orbits about the planet.
The planet and each of its moons may be considered to be point masses.

Show that the radius x of a moon's orbit is related to the period T of the orbit by the expression

$$GM = \frac{4\pi^2 x^3}{T^2}$$

where G is the gravitational constant and M is the mass of the planet. Explain your working.

[3]

(c) The planet Neptune has eight moons, each in a circular orbit of radius x and period T. The variation with T^2 of x^3 for some of the moons is shown in Fig. 1.1.

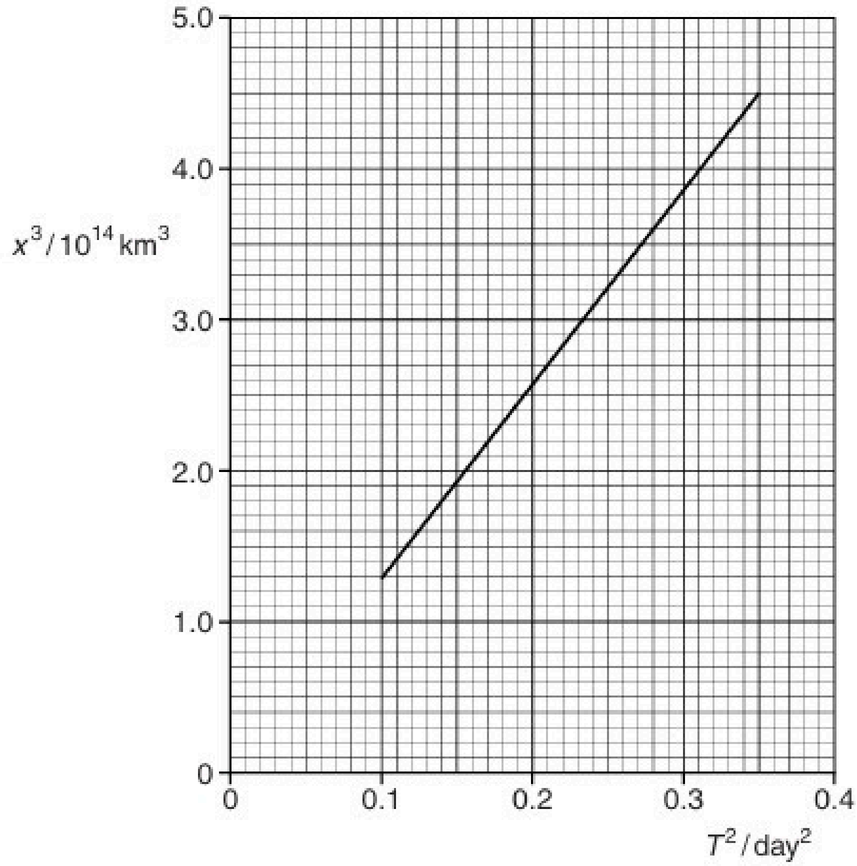


Fig. 1.1

Use Fig. 1.1 and the expression in **(b)** to determine the mass of Neptune.

mass = ... kg [4]

22. F/M/16/42/Q1

(a) State Newton's law of gravitation.

....

....

....

....[2]

(b) A satellite of mass m has a circular orbit of radius r about a planet of mass M. It may be assumed that the planet and the satellite are uniform spheres that are isolated in space.

Show that the linear speed v of the satellite is given by the expression

$$v = \sqrt{\frac{GM}{r}}$$

where G is the gravitational constant.
Explain your working.

[2]

(c) Two moons A and B have circular orbits about a planet, as illustrated in Fig. 1.1.

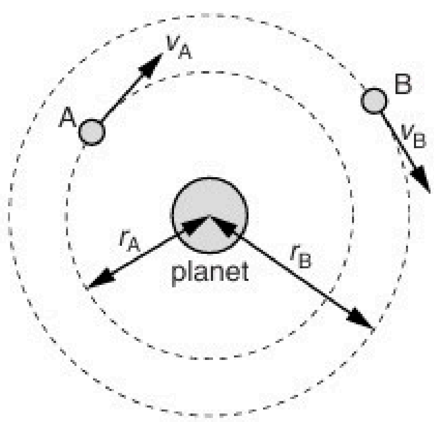


Fig. 1.1 (not to scale)

Moon A has an orbital radius r_A of 1.3×10^8 m, linear speed v_A and orbital period T_A.
Moon B has an orbital radius r_B of 2.2×10^{10} m, linear speed v_B and orbital period T_B.

(i) Determine the ratio

1. $\dfrac{v_A}{v_B}$,

ratio = ...[2]

2. $\dfrac{T_A}{T_B}$.

ratio = ...[3]

(ii) The planet spins about its own axis with angular speed $1.7 \times 10^{-4}\,\text{rad}\,\text{s}^{-1}$.
Moon A is always above the same point on the planet's surface.

Determine the orbital period T_B of moon B.

T_B = ... s [2]

[Total: 11]

23. M/J/16/41/Q1

(a) By reference to the definition of gravitational potential, explain why gravitational potential is a negative quantity.

....

....

.... [2]

(b) Two stars A and B have their surfaces separated by a distance of 1.4×10^{12} m, as illustrated in Fig. 1.1.

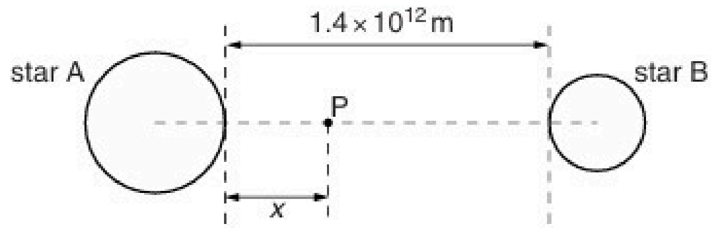


Fig. 1.1

Point P lies on the line joining the centres of the two stars. The distance x of point P from the surface of star A may be varied.

The variation with distance x of the gravitational potential ϕ at point P is shown in Fig. 1.2.

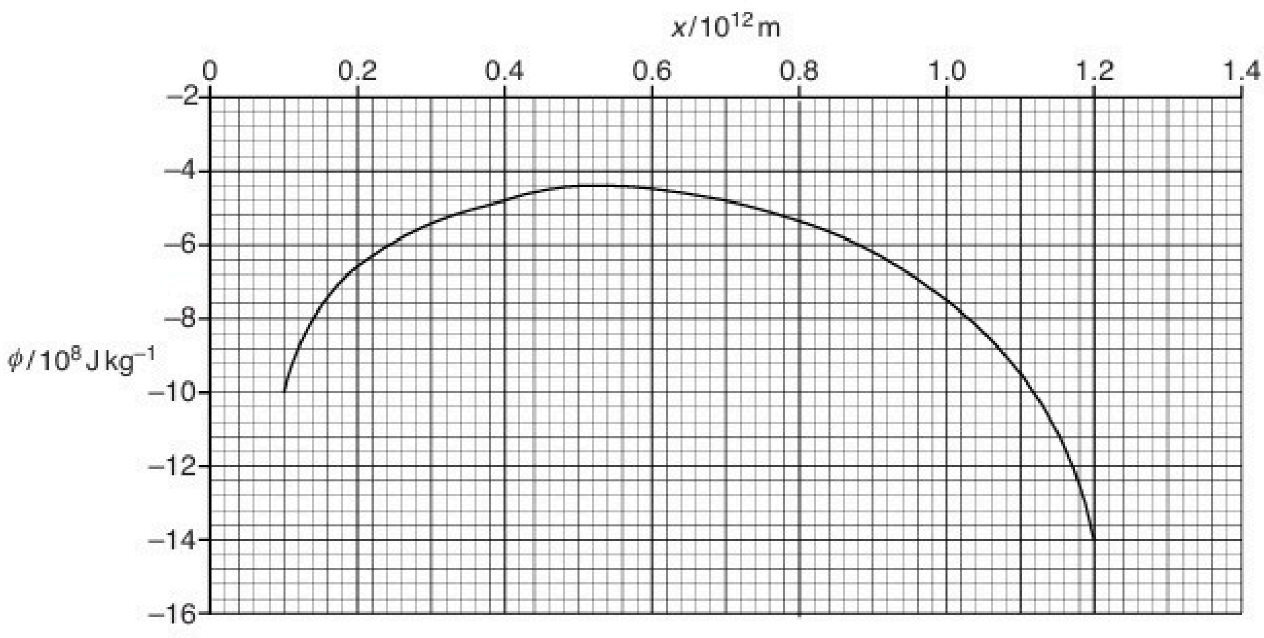


Fig. 1.2

A rock of mass 180 kg moves along the line joining the centres of the two stars, from star A towards star B.

(i) Use data from Fig. 1.2 to calculate the change in kinetic energy of the rock when it moves from the point where $x = 0.1 \times 10^{12}$ m to the point where $x = 1.2 \times 10^{12}$ m.
State whether this change is an increase or a decrease.

change = ... J

....
[3]

(ii) At a point where $x = 0.1 \times 10^{12}$ m, the speed of the rock is v.

Determine the minimum speed v such that the rock reaches the point where $x = 1.2 \times 10^{12}$ m.

minimum speed = ... $m\,s^{-1}$ [3]

[Total: 8]

24. M/J/16/42/Q1

A binary star consists of two stars A and B that orbit one another, as illustrated in Fig. 1.1.

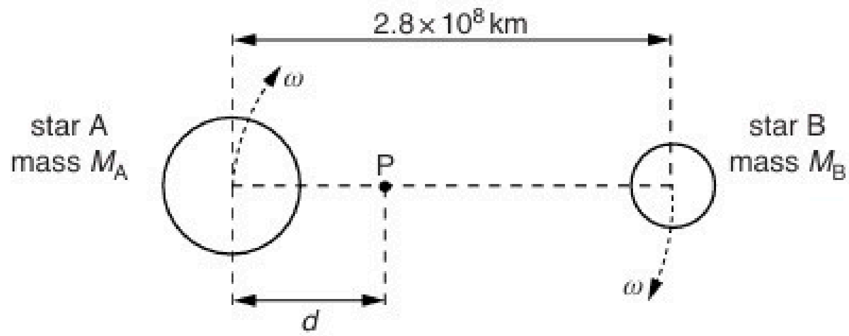


Fig. 1.1

The stars are in circular orbits with the centres of both orbits at point P, a distance d from the centre of star A.

(a) **(i)** Explain why the centripetal force acting on both stars has the same magnitude.

....

....

.... [2]

(ii) The period of the orbit of the stars about point P is 4.0 years.

Calculate the angular speed ω of the stars.

ω = ... $rad\,s^{-1}$ [2]

(b) The separation of the centres of the stars is 2.8×10^8 km.
The mass of star A is M_A. The mass of star B is M_B.
The ratio $\dfrac{M_A}{M_B}$ is 3.0.

(i) Determine the distance d.

d = ... km [3]

(ii) Use your answers in **(a)(ii)** and **(b)(i)** to determine the mass M_B of star B.
Explain your working.

M_B = ... kg [3]

[Total: 10]

25. O/N/16/41/Q1

A satellite is in a circular orbit of radius r about the Earth of mass M, as illustrated in Fig. 1.1.

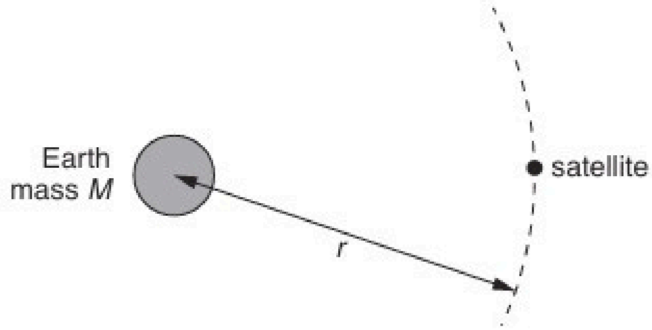


Fig. 1.1

The mass of the Earth may be assumed to be concentrated at its centre.

(a) Show that the period T of the orbit of the satellite is given by the expression

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

where G is the gravitational constant. Explain your working.

[3]

(b) (i) A satellite in geostationary orbit appears to remain above the same point on the Earth and has a period of 24 hours.
State two other features of a *geostationary* orbit.

1. ....

....

2. ....

....
[2]

(ii) The mass M of the Earth is 6.0×10^{24} kg.
Use the expression in **(a)** to determine the radius of a geostationary orbit.

radius = ... m [2]

(c) A global positioning system (GPS) satellite orbits the Earth at a height of 2.0×10^4 km above the Earth's surface.
The radius of the Earth is 6.4×10^3 km.

Use your answer in **(b)(ii)** and the expression

$$T^2 \propto r^3$$

to calculate, in hours, the period of the orbit of this satellite.

period = ... hours [2]

[Total: 9]

26. O/N/16/42/Q1

(a) Define *gravitational field strength*.

....

.... [1]

(b) The nearest star to the Sun is Proxima Centauri.
This star has a mass of 2.5×10^{29} kg and is a distance of 4.0×10^{13} km from the Sun.
The Sun has a mass of 2.0×10^{30} kg.

(i) State why Proxima Centauri may be assumed to be a point mass when viewed from the Sun.

....

.... [1]

(ii) Calculate

1. the gravitational field strength due to Proxima Centauri at a distance of 4.0×10^{13} km,

field strength = ... $N\,kg^{-1}$ [2]

2. the gravitational force of attraction between the Sun and Proxima Centauri.

force = ... N [2]

(c) Suggest quantitatively why it may be assumed that the Sun is isolated in space from other stars.

....

....

....[2]

[Total: 8]

27. M/J/17/41/Q1

(a) Explain how a satellite may be in a circular orbit around a planet.

....

....

....[2]

(b) The Earth and the Moon may be considered to be uniform spheres that are isolated in space. The Earth has radius R and mean density ρ. The Moon, mass m, is in a circular orbit about the Earth with radius nR, as illustrated in Fig. 1.1.

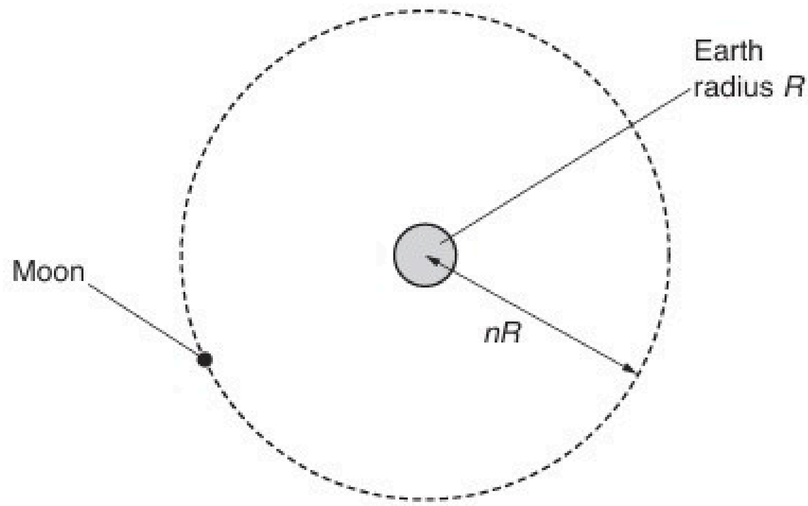


Fig. 1.1

The Moon makes one complete orbit of the Earth in time T.
Show that the mean density ρ of the Earth is given by the expression

$$\rho = \frac{3\pi n^3}{GT^2}.$$

[4]

(c) The radius R of the Earth is 6.38×10^3 km and the distance between the centre of the Earth and the centre of the Moon is 3.84×10^5 km.
The period T of the orbit of the Moon about the Earth is 27.3 days.
Use the expression in **(b)** to calculate ρ.

ρ = ... kg m^{-3} [3]

[Total: 9]

28. M/J/17/42/Q1

(a) Define *gravitational field strength*.

....

....[1]

(b) The mass of a spherical comet of radius 3.6 km is approximately 1.0×10^{13} kg.

(i) Assuming that the comet has constant density, calculate the gravitational field strength on the surface of the comet.

field strength = ... $N\,kg^{-1}$ [2]

(ii) A probe having a weight of 960 N on Earth lands on the comet.
Using your answer in **(i)**, determine the weight of the probe on the surface of the comet.

weight = ... N [2]

(c) A second comet has a length of approximately 4.5 km and a width of approximately 2.6 km. Its outline is illustrated in Fig. 1.1.

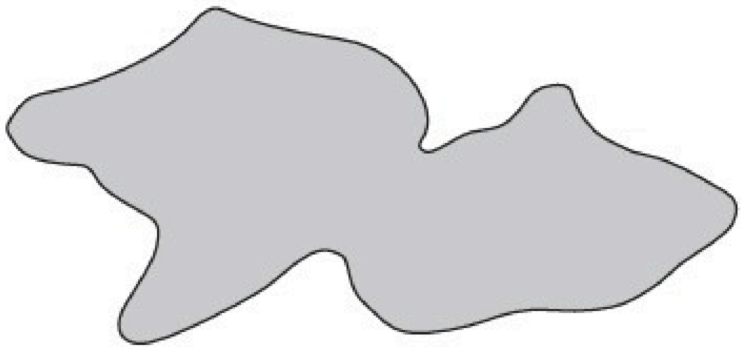


Fig. 1.1

Suggest one similarity and one difference between the gravitational fields at the surface of this comet and at the surface of the comet in **(b)**.

similarity: ....

....

difference: ....

....
[2]

[Total: 7]

29. O/N/17/41/Q3

(a) Define *gravitational field strength*.

....

.... [1]

(b) Explain why, for changes in vertical position of a point mass near the Earth's surface, the gravitational field strength may be considered to be constant.

....

....

....

.... [2]

(c) The orbit of the Earth about the Sun is approximately circular with a radius of 1.5×10^8 km. The time period of the orbit is 365 days.

Determine a value for the mass *M* of the Sun. Explain your working.

M = ... kg [5]

[Total: 8]

30. O/N/17/42/Q1

(a) State Newton's law of gravitation.

....

....

....[2]

(b) The planet Jupiter and one of its moons, Io, may be considered to be uniform spheres that are isolated in space.
Jupiter has radius R and mean density ρ.
Io has mass m and is in a circular orbit about Jupiter with radius nR, as illustrated in Fig. 1.1.

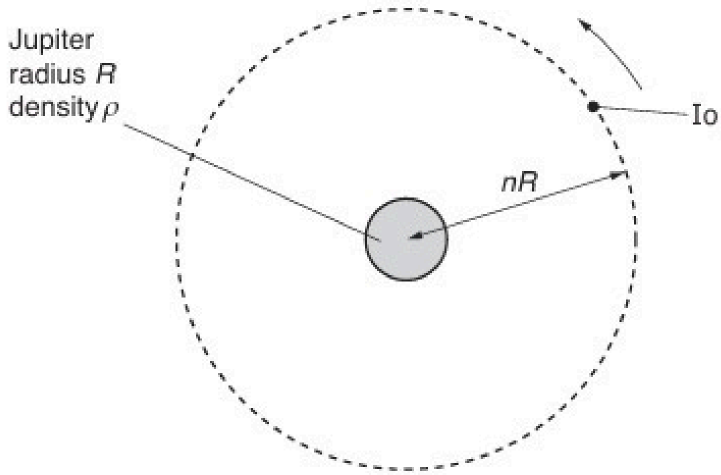


Fig. 1.1

The time for Io to complete one orbit of Jupiter is T.

Show that the time T is related to the mean density ρ of Jupiter by the expression

$$\rho T^2 = \frac{3\pi n^3}{G}$$

where G is the gravitational constant.

[4]

31. F/M/18/42/Q1

(a) (i) State what is meant by a line of force in a gravitational field.

....

....

....[1]

(ii) By reference to the pattern of the lines of gravitational force near to the surface of the Earth, explain why the acceleration of free fall near to the Earth's surface is approximately constant.

....

....

....

....

....[3]

(b) The Moon may be considered to be a uniform sphere that is isolated in space. It has radius 1.74×10^{3} km and mass 7.35×10^{22} kg.

(i) Calculate the gravitational field strength at the Moon's surface.

gravitational field strength = ... N kg^{-1} [2]

(ii) A satellite is in a circular orbit about the Moon at a height of 320 km above its surface.

Calculate the time for the satellite to complete one orbit of the Moon.

time = ... s [3]

_32. M/J/18/42/Q1_

(a) **(i)** A gravitational field may be represented by lines of gravitational force.
State what is meant by a *line of gravitational force*.

....

....

....[1]

(ii) By reference to lines of gravitational force near to the surface of the Earth, explain why the gravitational field strength g close to the Earth's surface is approximately constant.

....

....

....

....

....

....[3]

(b) The Moon may be considered to be a uniform sphere of diameter 3.4×10^3 km and mass 7.4×10^{22} kg. The Moon has no atmosphere.

During a collision of the Moon with a meteorite, a rock is thrown vertically up from the surface of the Moon with a speed of $2.8\,\text{km s}^{-1}$.

Assuming that the Moon is isolated in space, determine whether the rock will travel out into distant space or return to the Moon's surface.

[4]

[Total: 8]

33. O/N/18/41/Q1

(a) **(i)** State what is meant by *gravitational potential* at a point.

....

....

....[2]

(ii) Suggest why, for small changes in height near the Earth's surface, gravitational potential is approximately constant.

....

....

....

....[2]

(b) The Moon may be considered to be a uniform sphere with a diameter of 3.5×10^3 km and a mass of 7.4×10^{22} kg.

A meteor strikes the Moon and, during the collision, a rock is sent off from the surface of the Moon with an initial speed *v*.

Assuming that the Moon is isolated in space, determine the minimum speed of the rock such that it does not return to the Moon's surface. Explain your working.

minimum speed = ... $m s^{-1}$ [3]

[Total: 7]

34. O/N/18/42/Q1

(a) (i) State what is meant by *gravitational field strength*.

....

....

....[1]

(ii) Explain why, at the surface of a planet, gravitational field strength is numerically equal to the acceleration of free fall.

....

....

....[1]

(b) An isolated uniform spherical planet has radius R.
The acceleration of free fall at the surface of the planet is g.

On Fig. 1.1, sketch a graph to show the variation of the acceleration of free fall with distance x from the centre of the planet for values of x in the range $x = R$ to $x = 4R$.

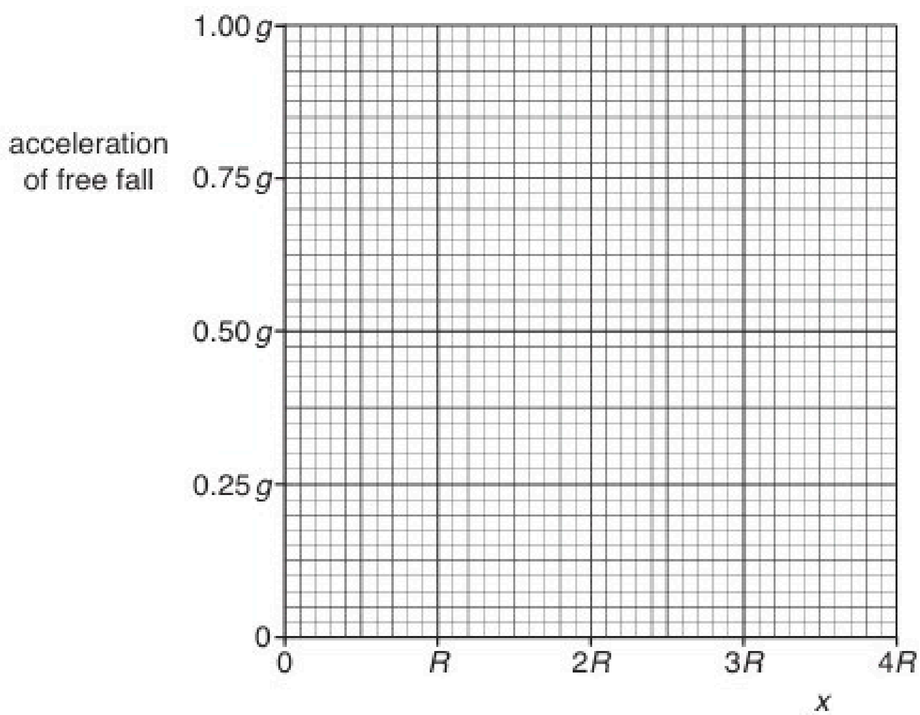


Fig. 1.1

[3]

(c) The planet in **(b)** has radius R equal to 3.4×10^3 km and mean density 4.0×10^3 kg m^{-3}.

Calculate the acceleration of free fall at a height R above its surface.

acceleration of free fall = ... m s^{-2} [3]

[Total: 8]

35. F/M/19/42/Q1

(a) (i) Define *gravitational potential* at a point.

....

....

.... [2]

(ii) Use your answer in **(i)** to explain why the gravitational potential near an isolated mass is always negative.

....

....

....

....

....

.... [3]

(b) A spherical planet has mass 6.00×10^{24} kg and radius 6.40×10^{6} m.
The planet may be assumed to be isolated in space with its mass concentrated at its centre.

A satellite of mass 340 kg is in a circular orbit about the planet at a height 9.00×10^{5} m above its surface.

For the satellite:

(i) show that its orbital speed is $7.4 \times 10^{3}\,\mathrm{m\,s^{-1}}$

[2]

(ii) calculate its gravitational potential energy.

energy = ... J [3]

(c) Rockets on the satellite are fired for a short time. The satellite's orbit is now closer to the surface of the planet.

State and explain the change, if any, in the kinetic energy of the satellite.

....

....

....

.... [2]

[Total: 12]

_36. M/J/19/41/Q1_

(a) Two point masses are isolated in space and are separated by a distance x.

State an expression relating the gravitational force F between the two masses to the magnitudes M and m of the masses. State the name of any other symbol used.

....

....

.... [1]

(b) A spacecraft is to be put into a circular orbit about a spherical planet.

The planet may be considered to be isolated in space. The mass of the planet, assumed to be concentrated at its centre, is 7.5×10^{23} kg. The radius of the planet is 3.4×10^{6} m.

(i) The spacecraft is to orbit the planet at a height of 2.4×10^{5} m above the surface of the planet. At this altitude, there is no atmosphere.

Show that the speed of the spacecraft in its orbit is $3.7 \times 10^{3}\,\mathrm{m\,s^{-1}}$.

[2]

(ii) One possible path of the spacecraft as it approaches the planet is shown in Fig. 1.1.

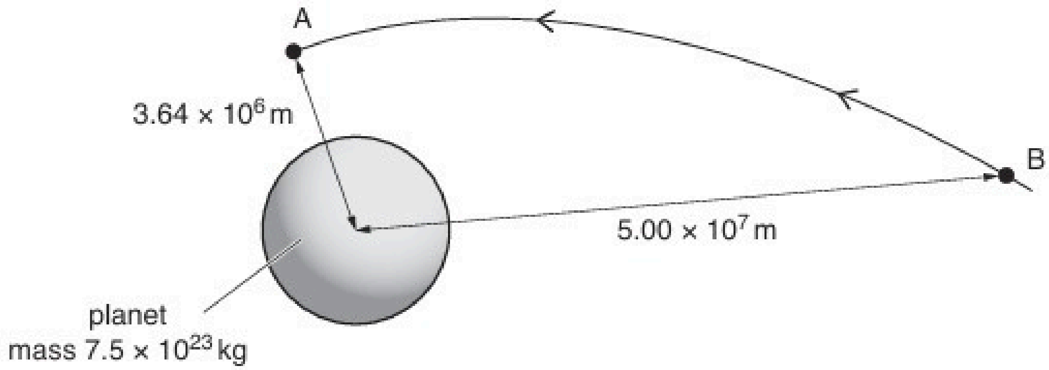


Fig. 1.1 (not to scale)

The spacecraft enters the orbit at point A with speed $3.7 \times 10^3\,m\,s^{-1}$.

At point B, a distance of 5.00×10^7 m from the centre of the planet, the spacecraft has a speed of $4.1 \times 10^3\,m\,s^{-1}$. The mass of the spacecraft is 650 kg.

For the spacecraft moving from point B to point A, show that the change in gravitational potential energy of the spacecraft is 8.3×10^9 J.

[3]

(c) By considering changes in gravitational potential energy and in kinetic energy of the spacecraft, determine whether the total energy of the spacecraft increases or decreases in moving from point B to point A. A numerical answer is not required.

....

....

....

.... [2]

37. M/J/19/42/Q1

(a) Two point masses are separated by a distance *x* in a vacuum.
State an expression for the force *F* between the two masses *M* and *m*.
State the name of any other symbol used.

....

....

....[1]

(b) A small sphere S is attached to one end of a rod, as shown in Fig. 1.1.

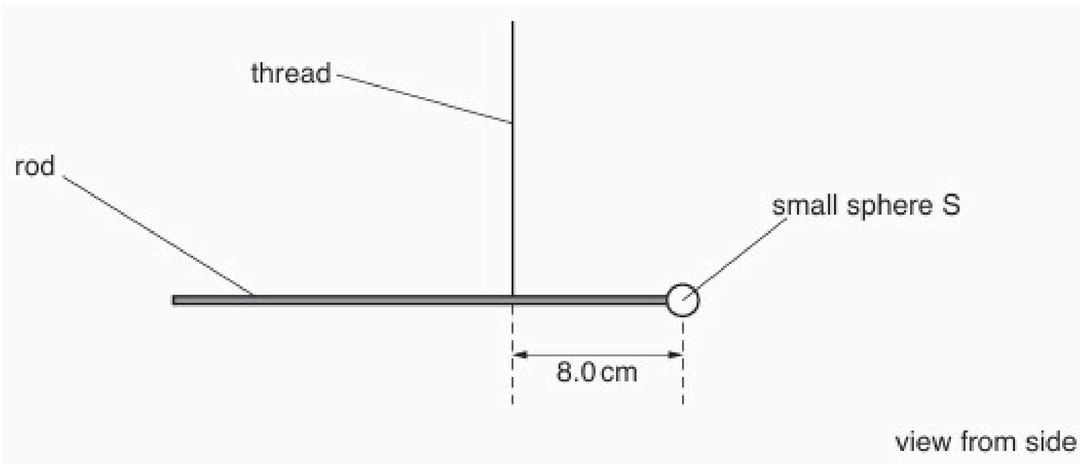


Fig. 1.1 (not to scale)

The rod hangs from a vertical thread and is horizontal.
The distance from the centre of sphere S to the thread is 8.0 cm.

A large sphere L is placed near to sphere S, as shown in Fig. 1.2.

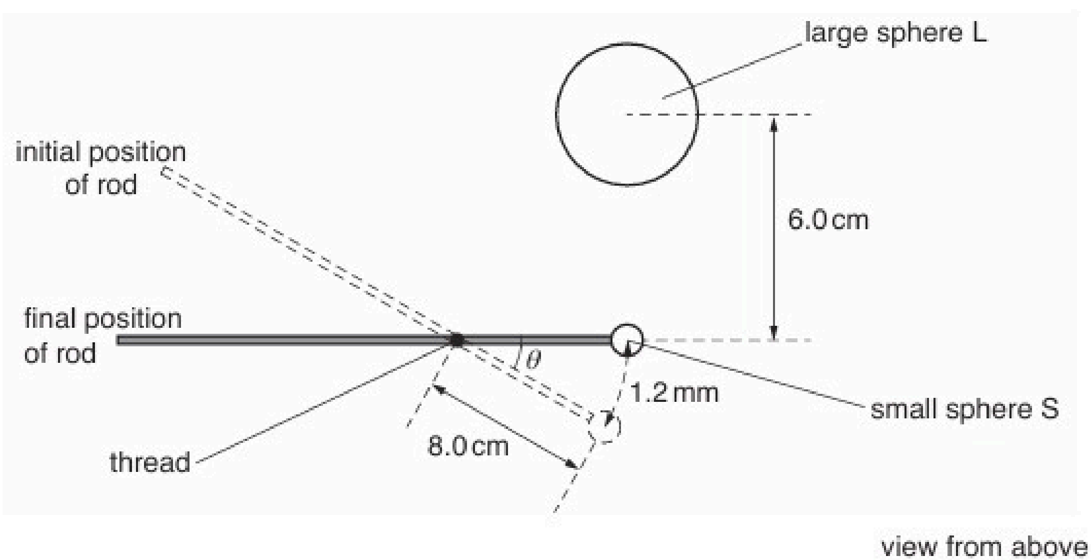


Fig. 1.2 (not to scale)

There is a force of attraction between spheres S and L, causing sphere S to move through a distance of 1.2 mm.
The line joining the centres of S and L is normal to the rod.

(i) Show that the angle θ through which the rod rotates is 1.5×10^{-2} rad.

[1]

(ii) The rotation of the rod causes the thread to twist.
The torque T (in N m) required to twist the thread through an angle β (in rad) is given by

$$T = 9.3 \times 10^{-10} \times \beta.$$

Calculate the torque in the thread when sphere L is positioned as shown in Fig. 1.2.

torque = ... N m [1]

(c) The distance between the centres of spheres S and L is 6.0 cm.
The mass of sphere S is 7.5 g and the mass of sphere L is 1.3 kg.

(i) By equating the torque in **(b)(ii)** to the moment about the thread produced by gravitational attraction between the spheres, calculate a value for the gravitational constant.

gravitational constant = ... N m^2 kg^{-2} [3]

(ii) Suggest why the total force between the spheres may not be equal to the force calculated using Newton's law of gravitation.

....

....[1]

[Total: 7]

38. O/N/19/41/Q1

(a) State Newton's law of gravitation.

....

....

.... [2]

(b) A geostationary satellite orbits the Earth. The orbit of the satellite is circular and the period of the orbit is 24 hours.

(i) State **two** other features of this orbit.

1. ....

....

2. ....

....
[2]

(ii) The radius of the orbit of the satellite is 4.23×10^4 km.

Determine a value for the mass of the Earth. Explain your working.

mass = ... kg [4]

[Total: 8]

39. O/N/19/42/Q1

(a) State Newton's law of gravitation.

....

....

.... [2]

(b) The astronomer Johannes Kepler showed that the period T of rotation of a planet about the Sun is related to its mean distance R from the centre of the Sun by the expression

$$\frac{R^3}{T^2} = k$$

where k is a constant.

Use Newton's law to show that, for planets in circular orbits about the Sun of mass M, the constant k is given by

$$k = \frac{GM}{4\pi^2}$$

where G is the gravitational constant. Explain your working.

[4]

(c) A satellite is in a circular orbit about Mars.
The radius of the orbit of the satellite is 4.38×10^6 m. The orbital period is 2.44 hours.

Use the expressions in **(b)** to calculate a value for the mass of Mars.

mass = ... kg [2]

[Total: 8]

40. F/M/20/42/Q1

(a) Define *gravitational potential* at a point.

....

....

.... [2]

(b) TESS is a satellite of mass 360 kg in a circular orbit about the Earth as shown in Fig. 1.1.

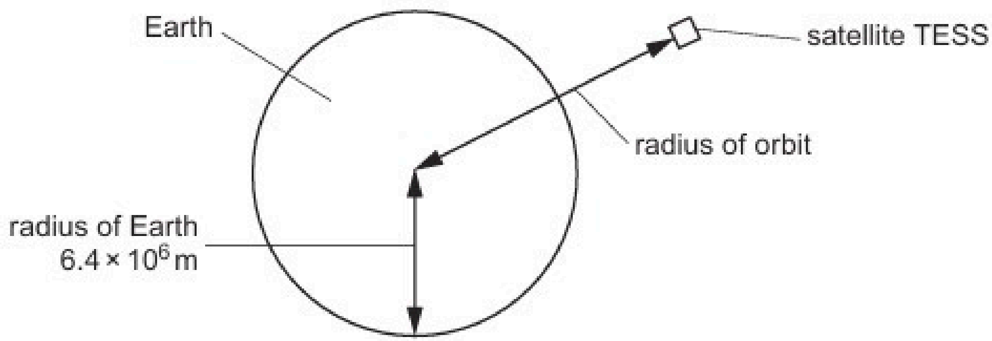


Fig. 1.1 (not to scale)

The radius of the Earth is 6.4×10^6 m and the mass of the Earth, considered to be a point mass at its centre, is 6.0×10^{24} kg.

(i) It takes TESS 13.7 days to orbit the Earth.

Show that the radius of orbit of TESS is 2.4×10^8 m.

[3]

(ii) Calculate the change in gravitational potential energy between TESS in orbit and TESS on a launch pad on the surface of the Earth.

change in gravitational potential energy = ... J [3]

(iii) Use the information in **(b)(i)** to calculate the ratio:

$$\frac{\text{gravitational field strength on surface of Earth}}{\text{gravitational field strength at location of TESS in orbit}}.$$

ratio = ... [2]

[Total: 10]

41. M/J/20/42/Q1

(a) Define *gravitational potential* at a point.

....

....

.... [2]

(b) An isolated solid sphere of radius r may be assumed to have its mass M concentrated at its centre. The magnitude of the gravitational potential at the surface of the sphere is ϕ.

On Fig. 1.1, show the variation of the gravitational potential with distance d from the centre of the sphere for values of d from $d = r$ to $d = 4r$.

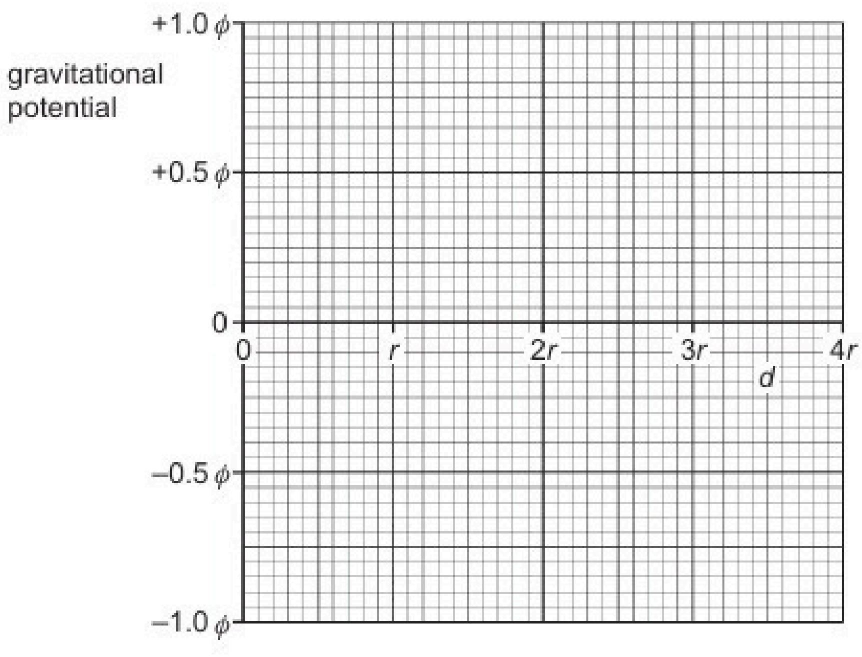


Fig. 1.1

[3]

(c) The sphere in **(b)** is a planet with radius r of 6.4×10^6 m and mass M of 6.0×10^{24} kg. The planet has no atmosphere.

A rock of mass 3.4×10^3 kg moves directly towards the planet. Its distance from the centre of the planet changes from $4r$ to $3r$.

(i) Calculate the change in gravitational potential energy of the rock.

change = ... J [3]

(ii) Explain whether the rock's speed increases, decreases or stays the same.

....

.... [2]

[Total: 10]

42. O/N/20/41/Q1

(a) **(i)** State what is meant by a *field of force*.

....

....

.... [2]

(ii) Define *gravitational field strength*.

....

.... [1]

(b) An isolated planet may be assumed to be a uniform sphere of radius 3.39×10^6 m with its mass of 6.42×10^{23} kg concentrated at its centre.

Calculate the gravitational field strength at the surface of the planet.

field strength = ... N kg^{-1} [3]

(c) Calculate the height above the surface of the planet in **(b)** at which the gravitational field strength is 1.0% less than its value at the surface of the planet.

height = ... m [3]

[Total: 9]

43. O/N/20/42/Q1

(a) Define *gravitational potential* at a point.

....

....

.... [2]

(b) The Earth may be considered to be a uniform sphere of radius 6.4×10^6 m with its mass of 6.0×10^{24} kg concentrated at its centre.

A satellite of mass 2.4×10^3 kg is launched from the Equator. It is placed in an equatorial orbit at a height of 5.6×10^6 m above the Earth's surface.

(i) Calculate the change ΔE_P in gravitational potential energy of the satellite for its movement from the surface of the Earth to its position in the equatorial orbit.

ΔE_P = ... J [3]

(ii) Determine the speed of the satellite when in orbit.

speed = ... $m s^{-1}$ [3]

(c) Before the satellite in **(b)** is launched, its speed at the Equator due to the Earth's rotation is $470\,\mathrm{m\,s^{-1}}$.

Suggest why the energy required to launch the satellite depends on whether the satellite, in its orbit, is travelling from west to east or from east to west.

....

.... [1]

[Total: 9]

44. F/M/21/42/Q1

(a) State Newton's law of gravitation.

....

....

.... [2]

(b) Planets have been observed orbiting a star in another solar system. Measurements are made of the orbital radius r and the time period T of each of these planets.

The variation with R^3 of T^2 is shown in Fig. 1.1.

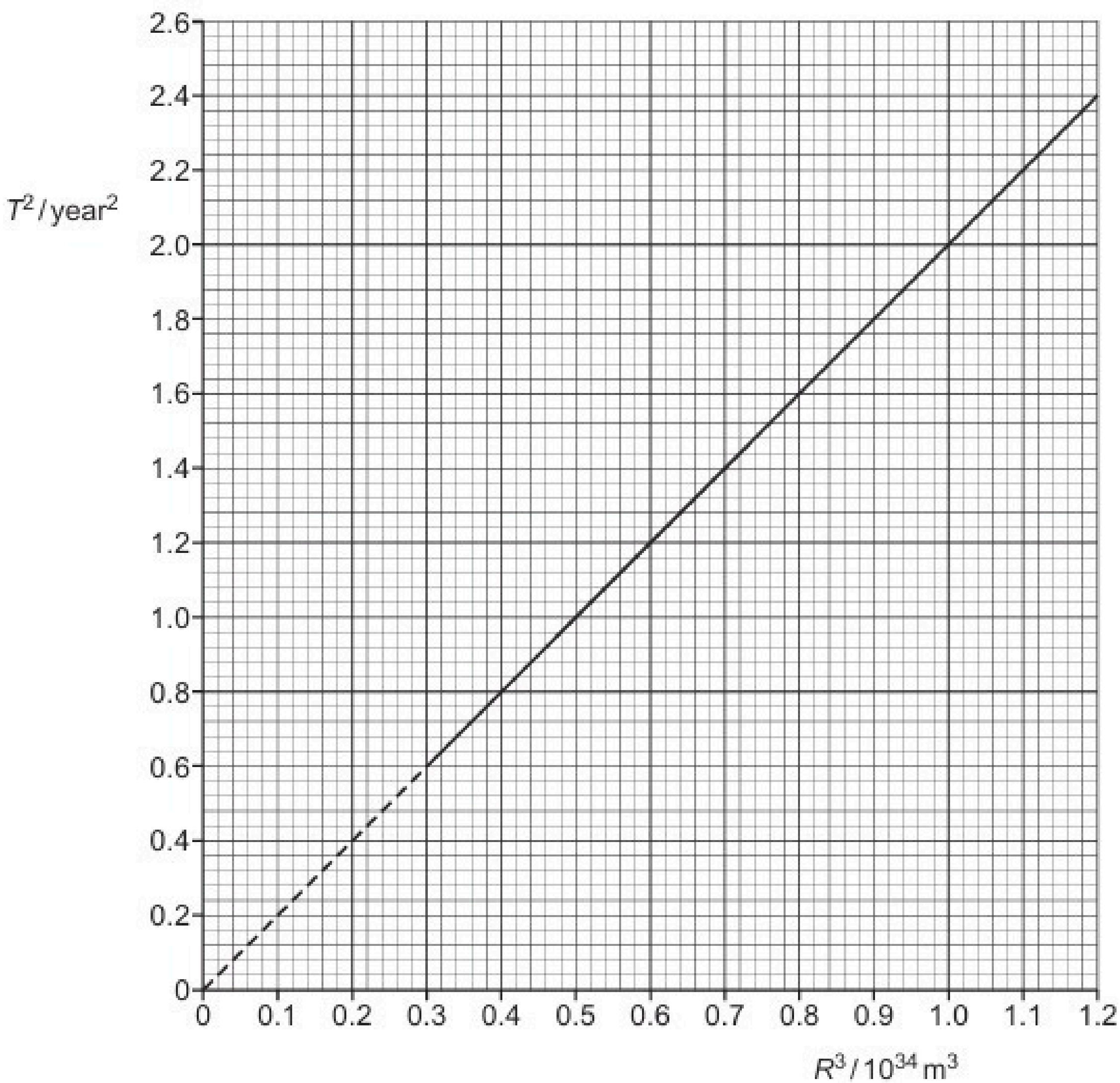


Fig. 1.1

The relationship between T and R is given by

$$T^2 = \frac{4\pi^2 R^3}{GM}$$

where G is the gravitational constant and M is the mass of the star.

Determine the mass M.

M = ... kg [3]

(c) A rock of mass m is also in orbit around the star in **(b)**. The radius of the orbit is r.

(i) Explain why the gravitational potential energy of the rock is negative.

....

....

....

.... [3]

(ii) Show that the kinetic energy E_k of the rock is given by

$$E_k = \frac{GMm}{2r}.$$

[2]

(iii) Use the expression in **(c)(ii)** to derive an expression for the total energy of the rock.

[2]

[Total: 12]

45. M/J/21/41/Q1

The Earth may be assumed to be an isolated uniform sphere with its mass of 6.0×10^{24} kg concentrated at its centre.

A satellite of mass 1200 kg is in a circular orbit about the Earth in the Earth's gravitational field. The period of the orbit is 94 minutes.

(a) Define *gravitational field strength*.

....

.... [1]

(b) Calculate the radius of the orbit of the satellite.

radius = ... m [3]

(c) Rockets on the satellite are fired so that the satellite enters a different circular orbit that has a period of 150 minutes. The change in the mass of the satellite may be assumed to be negligible.

(i) Show that the radius of the new orbit is 9.4×10^6 m.

[2]

(ii) State, with a reason, whether the gravitational potential energy of the satellite increases or decreases.

....

.... [1]

(iii) Determine the magnitude of the change in the gravitational potential energy of the satellite.

change in potential energy = ... J [3]

[Total: 10]

46. M/J/21/42/Q1

(a) Define *gravitational field strength*.

....

.... [1]

(b) An isolated planet is a uniform sphere of radius 3.39×10^{6} m. Its mass of 6.42×10^{23} kg may be considered to be a point mass concentrated at its centre. The planet rotates about its axis with a period of 24.6 hours.

For an object resting on the surface of the planet at the equator, calculate, to three significant figures:

(i) the gravitational field strength

field strength = ... $N\,kg^{-1}$ [2]

(ii) the centripetal acceleration

acceleration = ... $m\,s^{-2}$ [2]

(iii) the force per unit mass exerted on the object by the surface of the planet.

force per unit mass = ... $N\,kg^{-1}$ [1]

[Total: 6]

47. F/M/22/42/Q1

(a) The point P in Fig. 1.1 represents a point mass.

On Fig. 1.1, draw lines to represent the gravitational field around P.

Fig. 1.1

[2]

(b) A moon is in circular orbit around a planet.

Explain why the path of the moon is circular.

....

....

....

.... [2]

(c) Many moons are in circular orbit about a planet.

The angular velocity of a moon is ω when the orbit of the moon has a radius r about the planet.

Fig. 1.2 shows the variation of r^3 with $1/\omega^2$ for these moons.

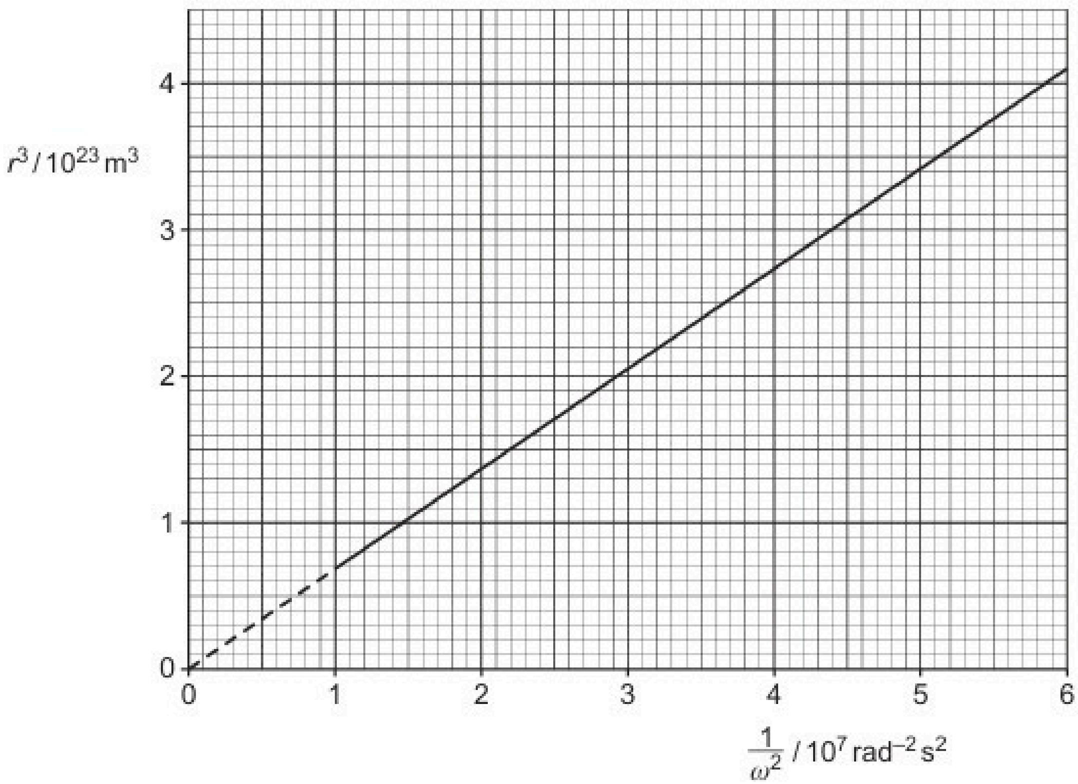


Fig. 1.2

(i) Show that the mass M of the planet is given by the expression

$$M = \frac{\text{gradient}}{G}$$

where G is the gravitational constant.

[2]

(ii) Use Fig. 1.2 and the expression in **(c)(i)** to show that the mass *M* of the planet is 1.0×10^{26} kg.

[1]

(iii) Determine the speed of a moon in orbit around the planet with an orbital radius of 1.2×10^{8} m.

speed = ... $\mathrm{m\,s^{-1}}$ [3]

[Total: 10]

48. M/J/22/41/Q1

(a) **(i)** State Newton's law of gravitation.

....

....

.... [2]

(ii) Use Newton's law of gravitation to show that the gravitational field strength g at a distance r away from a point mass M is given by

$$g = \frac{GM}{r^2}.$$

[2]

(b) The Earth has a mass of 5.98×10^{24} kg and a radius of 6.37×10^6 m.
The Moon has a mass of 7.35×10^{22} kg and a radius of 1.74×10^6 m.
The Earth and the Moon can both be considered as point masses at their centres. Their centres are a distance of 3.84×10^8 m apart.

(i) Show that the gravitational field strength at the surface of the Moon due to the mass of the Moon is $1.62\,\text{N kg}^{-1}$.

[1]

(ii) Explain why there is a point X on the line between the centres of the Earth and the Moon where the resultant gravitational field strength due to the Earth and the Moon is zero.

....

....

.... [2]

(iii) Calculate the distance x of point X from the centre of the Moon.

x = ... m [3]

[Total: 10]

49. M/J/22/42/Q1

(a) **(i)** Define gravitational potential at a point.

....

....

.... [2]

(ii) Starting from the equation for the gravitational potential due to a point mass, show that the gravitational potential energy E_P of a point mass m at a distance r from another point mass M is given by

$$E_P = -\frac{GMm}{r}$$

where G is the gravitational constant.

[1]

(b) Fig. 1.1 shows the path of a comet of mass 2.20×10^{14} kg as it passes around a star of mass 1.99×10^{30} kg.

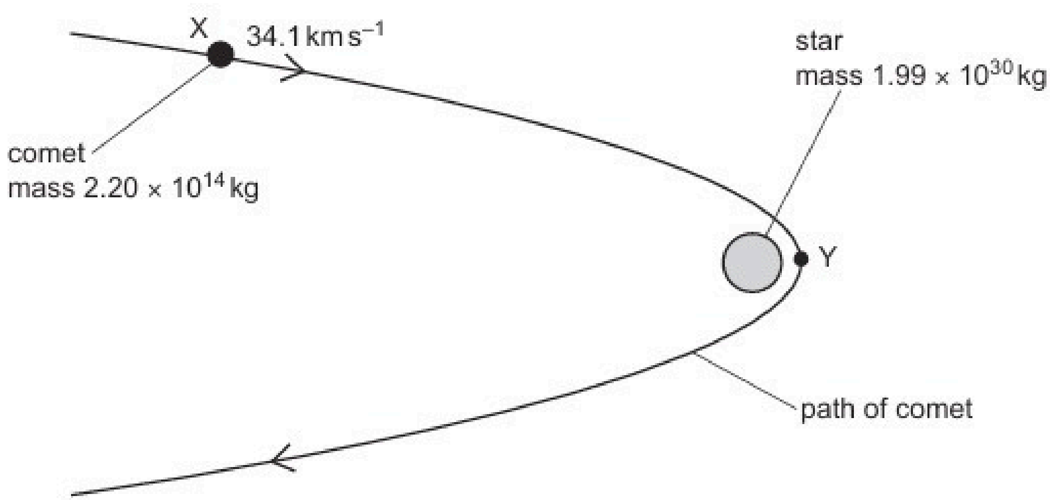


Fig. 1.1 (not to scale)

At point X, the comet is 8.44×10^{11} m from the centre of the star and is moving at a speed of $34.1\,\text{km s}^{-1}$.

At point Y, the comet passes its point of closest approach to the star. At this point, the comet is a distance of 6.38×10^{10} m from the centre of the star.

Both the comet and the star can be considered as point masses at their centres.

(i) Calculate the magnitude of the change in the gravitational potential energy ΔE_P of the comet as it moves from position X to position Y.

ΔE_P = ... J [2]

(ii) State, with a reason, whether the change in gravitational potential energy in **(b)(i)** is an increase or a decrease.

....

.... [1]

(iii) Use your answer in **(b)(i)** to determine the speed, in $km\,s^{-1}$, of the comet at point Y.

speed = ... $km\,s^{-1}$ [3]

(c) A second comet passes point X with the same speed as the comet in **(b)** and travelling in the same direction. This comet is gradually losing mass. The mass of this comet when it passes point X is the same as the mass of the comet in **(b)**.

Suggest, with a reason, how the path of the second comet compares with the path shown in Fig. 1.1.

....

.... [1]

50. O/N/22/41/Q1

(a) State the equation for the gravitational force F between two point masses m_1 and m_2 that are separated by a distance r. State the meaning of any other symbols you use.

[2]

(b) A satellite is in a circular orbit of radius R around a planet of mass M.

Show that the period T of the orbit is given by

$$T^2 = kR^3$$

where k is a constant that depends on the value of M. Explain your reasoning.

[3]

(c) A satellite is in a circular orbit around the Earth with a period of 24 hours.
The mass of the Earth is 6.0×10^{24} kg.

(i) Calculate the radius of the orbit.

radius = ... m [2]

(ii) State the **two** other conditions that must be met for the orbit to be geostationary.

1 ....

....

2 ....

....

[2]

[Total: 9]

51. O/N/22/42/Q1

(a) Define gravitational field.

....

.... [1]

(b) A spherical planet can be considered as a point mass at its centre.

(i) On Fig. 1.1, draw gravitational field lines outside the planet to represent the gravitational field due to the planet.

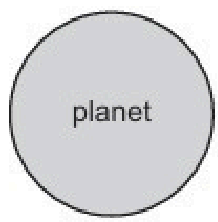


Fig. 1.1

[2]

(ii) A satellite is in a circular orbit around the planet.

Explain, with reference to your answer in **(b)(i)**, why the path of the satellite is circular.

....

....

.... [2]

(c) An object rests on the surface of the Earth at the Equator.
The radius of the Earth is 6.4×10^6 m.

(i) Determine the centripetal acceleration of the object.

centripetal acceleration = ... $m s^{-2}$ [3]

(ii) Describe how the two forces acting on the object give rise to this centripetal acceleration.
You may draw a diagram if you wish.

....

....

.... [2]

[Total: 10]

52. F/M/23/42/Q1

(a) Define gravitational potential at a point.

....

....

.... [2]

(b) Artemis is a spherical planet that may be assumed to be isolated in space. The variation with distance x from the centre of Artemis of the gravitational potential ϕ is shown in Fig. 1.1.

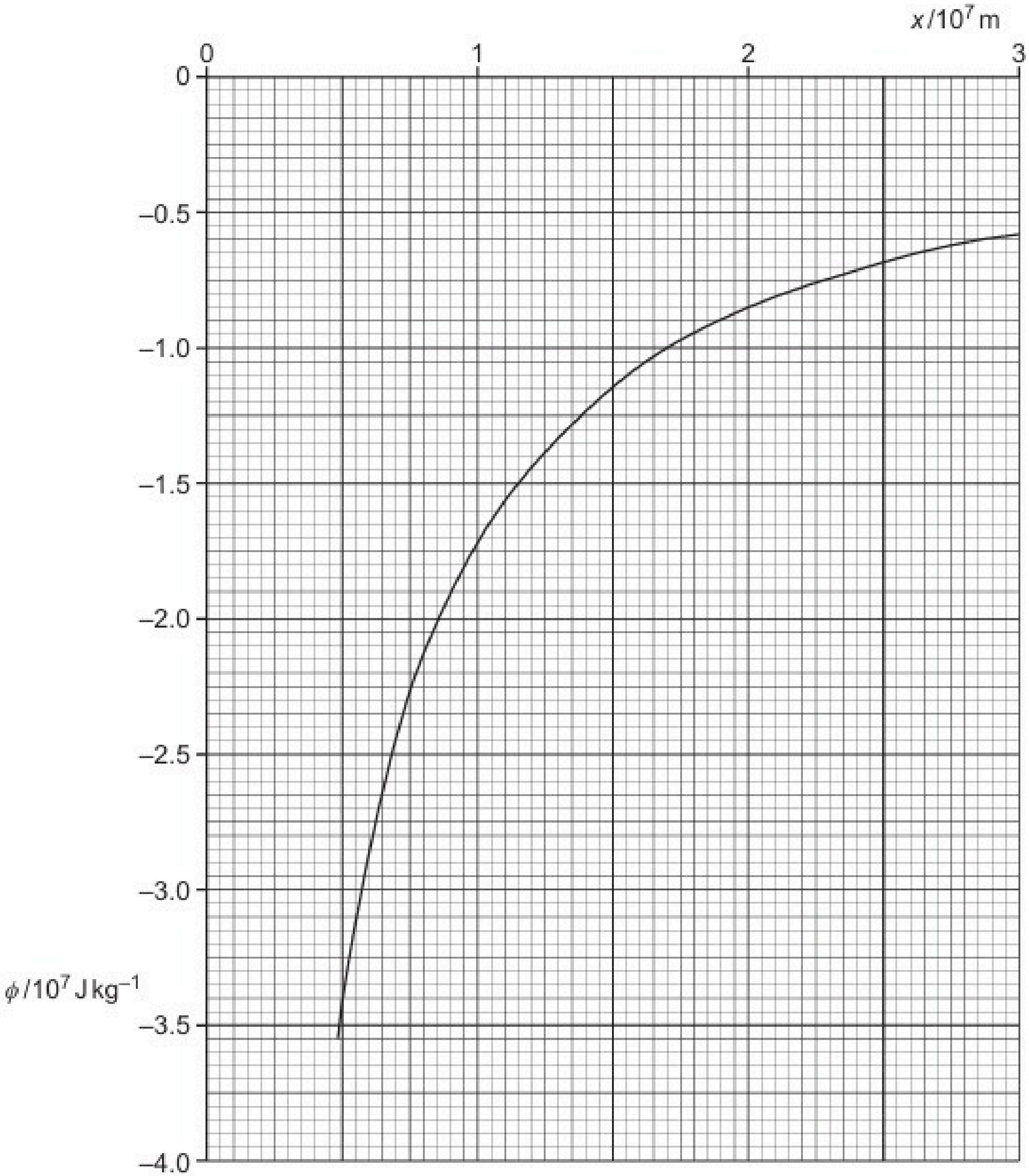


Fig. 1.1

(i) The radius of Artemis is 4800 km.

Determine the value of ϕ on the surface of Artemis.

ϕ = ... $\mathrm{J\,kg^{-1}}$ [1]

(ii) Show that the mass of Artemis is 2.55×10^{24} kg.

[1]

(iii) Calculate the gravitational field strength g on the surface of Artemis.

g = ... $\mathrm{N\,kg^{-1}}$ [2]

(iv) A satellite is in an orbit at a fixed position above a point on the surface of Artemis. The satellite is located above the equator of Artemis at a height above the surface where the gravitational potential is $-0.65 \times 10^{7}\,\mathrm{J\,kg^{-1}}$.

Calculate the period, in hours, of rotation of Artemis.

period = ... hours [4]

(c) State **one** similarity and **one** difference between gravitational potential due to a point mass and electric potential due to a point charge.

similarity ....

....

difference ....

....
[2]

[Total: 12]

53. M/J/23/41/Q1

(a) **(i)** Define gravitational field.

....

.... [1]

(ii) Define electric field.

....

.... [1]

(iii) State **one** similarity and **one** difference between the gravitational potential due to a point mass and the electric potential due to a point charge.

similarity: ....

....

difference: ....

....

[2]

(b) An isolated uniform conducting sphere has mass M and charge Q.
The gravitational field strength at the surface of the sphere is g.
The electric field strength at the surface of the sphere is E.

(i) Show that

$$\frac{M}{Q} = \alpha \frac{g}{E}$$

where α is a constant.

[3]

(ii) Show that the numerical value of α is $1.35 \times 10^{20}\,\mathrm{kg^2\,C^{-2}}$.

[1]

(c) Assume that the Earth is a uniform conducting sphere of mass 5.98×10^{24} kg.
The surface of the Earth carries a charge of -4.80×10^{5} C that is evenly distributed.

(i) Use the information in **(b)** to determine the electric field strength at the surface of the Earth. Give a unit with your answer.

electric field strength = ... unit .... [2]

(ii) State how the direction of the electric field at the surface of the Earth compares with the direction of the gravitational field.

.... [1]

[Total: 11]

54. M/J/23/42/Q1

(a) State Newton's law of gravitation.

....

....

.... [2]

(b) A satellite is in a circular orbit around a planet. The radius of the orbit is R and the period of the orbit is T. The planet is a uniform sphere.

Use Newton's law of gravitation to show that R and T are related by

$$4\pi^2 R^3 = GMT^2$$

where M is the mass of the planet and G is the gravitational constant.

[2]

(c) The Earth may be considered to be a uniform sphere of mass 5.98×10^{24} kg and radius 6.37×10^6 m.

A geostationary satellite is in orbit around the Earth.

Use the expression in **(b)** to determine the height of the satellite above the Earth's surface.

height = ... m [3]

(d) Another satellite is in a circular orbit around the Earth with the same orbital radius and period as the satellite in **(c)**.

(i) Calculate the angular speed of the satellite in this orbit. Give a unit with your answer.

angular speed = ... unit .... [2]

(ii) Despite having the same orbital period, the orbit of this satellite is not geostationary.

Suggest **two** ways in which the orbit of this satellite could be different from the orbit of the satellite in **(c)**.

1 ....

....

2 ....

....
[2]

[Total: 11]

55. O/N/23/41/Q1

(a) (i) State what is indicated by the direction of the gravitational field line at a point in a gravitational field.

....

.... [1]

(ii) Explain, with reference to gravitational field lines, why the gravitational field near the surface of the Earth is approximately constant for small changes in height.

....

....

.... [2]

(b) A large isolated uniform sphere has mass M and radius R.

Point P lies on a straight line passing through the centre of the sphere, at a variable displacement x from the centre, as shown in Fig. 1.1.

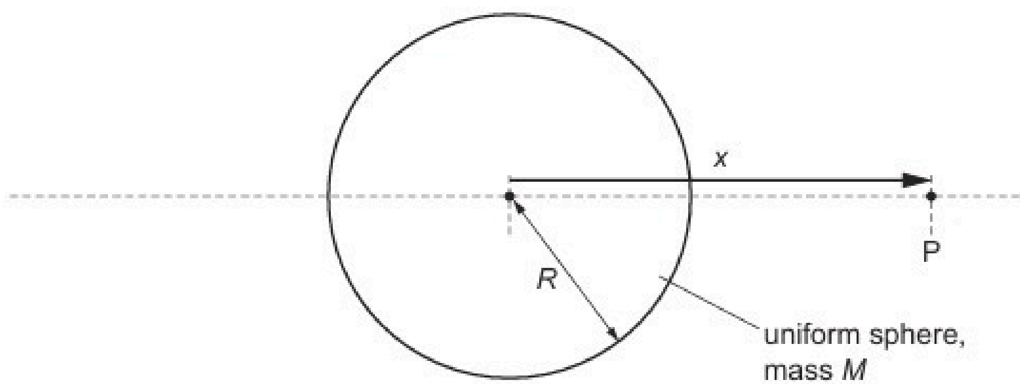


Fig. 1.1

Fig. 1.2 shows the variation with x of the gravitational field g at point P due to the sphere for the values of x for which P is inside the sphere.

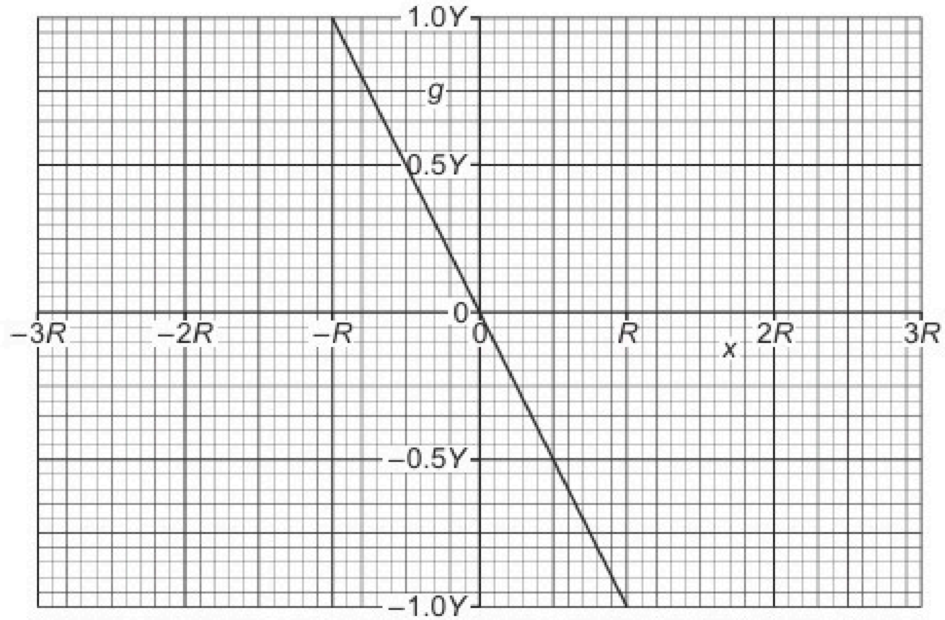


Fig. 1.2

The magnitude of the gravitational field at the surface of the sphere is Y.

(i) Determine an expression for Y in terms of M and R. Identify any other symbols that you use.

[2]

(ii) Explain why, at the surface of the sphere, g always has the opposite sign to x.

....

....

.... [2]

(iii) Complete Fig. 1.2 to show the variation of g with x for values of x, up to $\pm 3R$, for which point P is outside the sphere. [3]

[Total: 10]

ANSWERS

1.

(a)		(region of space) where a <u>mass</u> experiences a force	B1
(b)	**(i)**	potential energy = (–)GMm / x	C1
		$\Delta E_P = GMm/2R - GMm/3R$	M1
		$= GMm/6R$	A0
	(ii)	$E_K = ½m\,(7600^2 - 7320^2)$	M1
		$= (2.09 \times 10^6)m$	A0
(c)	**(i)**	$2.09 \times 10^6 = (6.67 \times 10^{-11}\,M)/(6 \times 3.4 \times 10^6)$	C1
		$M = 6.39 \times 10^{23}$ kg	A1
	(ii)	e.g. no energy dissipated due to friction with <u>atmosphere/air</u>	
		rocket is outside atmosphere	
		not influenced by another planet etc.	B1

2.

(a)		force per unit mass *(ratio idea essential)*	B1
(b)		$g = GM / R^2$	C1
		$8.6 \times (0.6 \times 10^7)^2 = M \times 6.67 \times 10^{-11}$	C1
		$M = 4.6 \times 10^{24}$ kg	A1
(c)	**(i)**	*either* potential decreases as distance from planet decreases	
		or potential zero at infinity and X is closer to zero	
		or potential $\alpha\ -1/r$ and Y more negative	M1
		so point Y is closer to planet.	A1
	(ii)	idea of $\Delta\phi = ½v^2$	C1
		$(6.8 - 5.3) \times 10^7 = ½v^2$	
		$v = 5.5 \times 10^3$ ms^{-1}	A1

3.

(a) (i) force per (unit) mass ……(*ratio idea essential*) B1

(ii) $g = GM / R^2$ C1

$9.81 = (6.67 \times 10^{-11} \times M) / (6.38 \times 10^6)^2$ ……(*all 3 s.f*) M1

$M = 5.99 \times 10^{24}$ kg A0

(b) (i) *either* $GM = \omega^2 r^3$ *or* $gR^2 = \omega^2 r^3$ C1

either $6.67 \times 10^{-11} \times 5.99 \times 10^{24} = \omega^2 \times (2.86 \times 10^7)^3$

or $9.81 \times (6.38 \times 10^6)^2 = \omega^2 \times (2.86 \times 10^7)^3$ C1

$\omega = 1.3 \times 10^{-4}$ rad s^{-1} A1

(*use of* $r = 2.22 \times 10^7$*m scores max 2 marks*)

(ii) period of orbit $= 2\pi / \omega$ C1

$= 4.8 \times 10^4$ s (= 13.4 hours) A1

period for geostationary satellite is 24 hours (= 8.6×10^4 s) A1

so no A0

(c) satellite can then provide cover at Poles B1

4.

(a) $F \propto Mm / R^2$ ……..(*words or explained symbols*) M1

either M and m are point masses

or R >> diameter of masses …(*do not allow 'size'*) A1

(b) (i) equatorial orbit B1

period 24 hours / same angular speed B1

from west to east / same direction of rotation B1

(*allow one of the last two marks for 'always overhead' if 2nd or 3rd marks not scored*)

(ii) gravitational force provides centripetal force

/ gives rise to centripetal acceleration ….(*in 'words'*) B1

$GM / x^2 = x\omega^2$ M1

$g = GM / R^2$ M1

to give $gR^2 = x^3\omega^2$ A0

(iii) $\omega = 2\pi / (24 \times 3600) = 7.27 \times 10^{-5}$ rad s^{-1} C1

$9.81 \times (6.4 \times 10^6)^2 = x^3 \times (7.27 \times 10^{-5})^2$ C1

$x^3 = 7.6 \times 10^{22}$

$x = 4.2 \times 10^7$ m A1

(*use of* $g = 10$ m s^{-2}*, loses 1 mark but once only in the Paper*)

5.

(a)	work done moving <u>unit</u> mass from infinity to the point	M1 A1
(b) (i)	at R, $\phi = 6.3 \times 10^7$ J kg^{-1} (allow $\pm 0.1 \times 10^7$) $\phi = GM / R$ $6.3 \times 10^7 = (6.67 \times 10^{-11} \times M) / (6.4 \times 10^6)$ $M = 6.0 \times 10^{24}$ kg (allow 5.95 $\rightarrow$ 6.14) Maximum of 2/3 for any value chosen for ϕ not at R	B1 C1 A1
(ii)	change in potential $= 2.1 \times 10^7$ J kg^{-1} (allow $\pm 0.1 \times 10^7$) loss in potential energy = gain in kinetic energy $\frac{1}{2} mv^2 = \phi m$ or $\frac{1}{2} mv^2 = GM / 3R$ $\frac{1}{2} v^2 = 2.1 \times 10^7$ $v = 6.5 \times 10^3$ m s^{-1} ……………..(allow 6.3 $\rightarrow$ 6.6) (answer 7.9×10^3 m s^{-1}, based on $x = 2R$, allow max 3 marks)	C1 B1 C1 A1
(iii)	e.g. speed / velocity / acceleration would be greater deviates / bends from straight path (any sensible ideas, 1 each, max 2)	B1 B1

6.

(a)	force per unit mass (*ratio idea essential*)	B1
(b)	graph: correct curvature from (R,1.0 g_S) & at least one other correct point	M1 A1
(c) (i)	fields of Earth and Moon are in opposite directions *either* resultant field found by subtraction of the field strength *or* any other sensible comment so there is a point where it is zero (*allow* $F_E = -F_M$ *for 2 marks*)	M1 A1 A0
(ii)	$GM_E / x^2 = GM_M / (D - x)^2$ $(6.0 \times 10^{24}) / (7.4 \times 10^{22}) = x^2 / (60R_E - x)^2$ $x = 54\,R_E$	C1 C1 A1
(iii)	graph: $g = 0$ at least ⅔ distance to Moon g_E and g_M in opposite directions correct curvature (by eye) and $g_E > g_M$ at surface	B1 M1 A1

7.

(a) (i) force proportional to product of masses B1
force inversely proportional to square of separation B1

(ii) separation <u>much</u> greater than radius / diameter of Sun / planet B1

(b) (i) e.g. force or field strength $\propto 1 / r^2$
potential $\propto 1 / r$ B1

(ii) e.g. gravitational force (always) attractive B1
electric force attractive or repulsive B1

8.

(a) work done in bringing unit mass from infinity (to the point) B1

(b) gravitational <u>force</u> is (always) attractive B1
either as r decreases, object/mass/body does work
or work is done by masses as they come together B1

(c) *either* force on mass = mg (where g is the acceleration of free fall /gravitational field strength) B1
$g = GM/r^2$ B1
if r @ h, g is constant B1
ΔE_P = force × distance moved M1
$= mgh$ A0
or $\Delta E_P = m\Delta\phi$ (C1
$= GMm(1/r_1 - 1/r_2) = GMm(r_2 - r_1)/r_1r_2$ (B1
if $r_2 \approx r_1$, then $(r_2 - r_1) = h$ and $r_1r_2 = r^2$ (B1
$g = GM/r^2$ (B1
$\Delta E_P = mgh$ (A0

(d) $\frac{1}{2}mv^2 = m\Delta\phi$
$v^2 = 2 \times GM/r$ C1
$= (2 \times 4.3 \times 10^{13}) / (3.4 \times 10^6)$ C1
$v = 5.0 \times 10^3\,\mathrm{m\,s^{-1}}$ A1
(Use of diameter instead of radius to give $v = 3.6 \times 10^3\,\mathrm{m\,s^{-1}}$ scores 2 marks)

9.

(a) force proportional to product of masses and inversely proportional to square of separation (*do not allow square of distance/radius*) M1

either point masses *or* separation @ size of masses A1

(b) (i) $\omega = 2\pi / (27.3 \times 24 \times 3600)$ or $2\pi / (2.36 \times 10^6)$ M1

$= 2.66 \times 10^{-6}$ rad s^{-1} A0

(ii) $GM = r^3\omega^2$ or $GM = v^2r$ C1

$M = (3.84 \times 10^5 \times 10^3)^3 \times (2.66 \times 10^{-6})^2 / (6.67 \times 10^{-11})$ M1

$= 6.0 \times 10^{24}$ kg A0

(special case: uses $g = GM/r^2$ with $g = 9.81$, $r = 6.4 \times 10^6$ scores max 1 mark

(c) (i) grav. force $= (6.0 \times 10^{24}) \times (7.4 \times 10^{22}) \times (6.67 \times 10^{-11})/(3.84 \times 10^8)^2$ C1

$= 2.0 \times 10^{20}$ N (*allow 1 SF*) A1

(ii) *either* $\Delta E_P = Fx$ because F constant as x ! radius of orbit B1

$\Delta E_P = 2.0 \times 10^{20} \times 4.0 \times 10^{-2}$ C1

$= 8.0 \times 10^{18}$ J (*allow 1 SF*) A1

or $\Delta E_P = GMm/r_1 - GMm/r_2$ C1

Correct substitution B1

8.0×10^{18} J A1

($\Delta E_P = GMm/r_1 + GMm/r_2$ is incorrect physics so 0/3)

10.

(a) force is proportional to the product of the masses and inversely proportional to the square of the separation M1

either point masses *or* separation >> size of masses A1

(b) (i) gravitational force provides the centripetal force B1

$mv^2/r = GMm/r^2$ and $E_K = ½mv^2$ M1

hence $E_K = GMm/2r$ A0

(ii) 1. $\Delta E_K = ½ \times 4.00 \times 10^{14} \times 620 \times (\{7.30 \times 10^6\}^{-1} - \{7.34 \times 10^6\}^{-1})$ C1

$= 9.26 \times 10^7$ J (*ignore any sign in answer*) A1

(*allow* 1.0×10^8 J *if evidence that* E_K *evaluated separately for each* r)

2. $\Delta E_P = 4.00 \times 10^{14} \times 620 \times (\{7.30 \times 10^6\}^{-1} - \{7.34 \times 10^6\}^{-1})$ C1

$= 1.85 \times 10^8$ J (*ignore any sign in answer*) A1

(*allow* 1.8 or 1.9×10^8 J)

(iii) *either* $(7.30 \times 10^6)^{-1} - (7.34 \times 10^6)^{-1}$ *or* ΔE_K is positive / E_K increased M1

speed has increased A1

11.

(a) region of space area / volume — B1
where a mass experiences a force — B1

(b) (i) force proportional to product of two masses — M1
force inversely proportional to the square of their separation — M1
either reference to point masses *or* separation >> 'size' of masses — A1

(ii) field strength $= GM / x^2$ **or** field strength $\propto 1 / x^2$ — C1
ratio $= (7.78 \times 10^8)^2 / (1.5 \times 10^8)^2$ — C1
$= 27$ — A1

(c) (i) *either* centripetal force $= mR\omega^2$ and $\omega = 2\pi / T$
or centripetal force $= mv^2 / R$ and $v = 2\pi R / T$ — B1
gravitational force provides the centripetal force — B1
either $GMm / R^2 = mR\omega^2$ *or* $GMm / R^2 = mv^2 / R$ — M1
$M = 4\pi^2 R^3 / GT^2$ — A0
(allow working to be given in terms of acceleration)

(ii) $M = \{4\pi^2 \times (1.5 \times 10^{11})^3\} / \{6.67 \times 10^{-11} \times (3.16 \times 10^7)^2\}$ — C1
$= 2.0 \times 10^{30}$ kg — A1

12.

(a) equatorial orbit / above equator — B1
satellite moves from west to east / same direction as Earth spins — B1
period is 24 hours / same period as spinning of Earth — B1
(allow 1 mark for 'appears to be stationary/overhead' if none of above marks scored)

(b) gravitational force provides/is the centripetal force — B1
$GMm/R^2 = mR\omega^2$ or $GMm/R^2 = mv^2/R$ — M1
$\omega = 2\pi / T$ or $v = 2\pi R / T$ or clear substitution — M1
clear working to give $R^3 = (GMT^2 / 4\pi^2)$ — A1

(c) $R^3 = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (24 \times 3600)^2 / 4\pi^2$ — C1
$= 7.57 \times 10^{22}$ — C1
$R = 4.2 \times 10^7$ m — A1
(missing out 3600 gives 1.8×10^5 m and scores 2/3 marks)

13.

(a) work done in moving unit mass — M1
from infinity (to the point) — A1

(b) (i) gravitational potential energy = GMm / x
energy = $(6.67 \times 10^{-11} \times 7.35 \times 10^{22} \times 4.5) / (1.74 \times 10^{6})$ — M1
energy = 1.27×10^{7} J — A0

(ii) <u>change in</u> grav. potential energy = <u>change in</u> kinetic energy — B1
$\frac{1}{2} \times 4.5 \times v^{2} = 1.27 \times 10^{7}$
$v = 2.4 \times 10^{3}$ m s^{-1} — A1

(c) Earth would attract the rock / potential at Earth('s surface) not zero / <0
/ at Earth, potential due to Moon not zero — M1
escape speed would be lower — A1

14.

(a) force proportional to product of the two masses and inversely proportional to the square of their separation — M1
either reference to point masses *or* separation >> 'size' of masses — A1

(b) gravitational force provides the centripetal force — B1
$GMm / R^{2} = mR\omega^{2}$ — M1
where *m is* the mass of the planet — A1
$GM = R^{3}\omega^{2}$ — A0

(c) $\omega = 2\pi / T$ — C1
either $M_{star} / M_{Sun} = (R_{star} / R_{Sun})^{3} \times (T_{Sun} / T_{star})^{2}$
$M_{star} = 4^{3} \times (\frac{1}{2})^{2} \times 2.0 \times 10^{30}$ — C1
$= 3.2 \times 10^{31}$ kg — A1
or $M_{star} = (2\pi)^{2} R_{star}{}^{3} / GT^{2}$ — (C1)
$= \{(2\pi)^{2} \times (6.0 \times 10^{11})^{3}\} / \{6.67 \times 10^{-11} \times (2 \times 365 \times 24 \times 3600)^{2}\}$ — (C1)
$= 3.2 \times 10^{31}$ kg — (A1)

15.

(a) work done bringing unit mass from infinity (to the point) — M1, A1

(b) $E_P = -m\phi$ — B1

(c) $\phi \propto 1/x$ — C1

either at 6R from centre, potential is $(6.3 \times 10^7)/6$ $(= 1.05 \times 10^7 \text{ J kg}^{-1})$
<u>and</u> at 5R from centre, potential is $(6.3 \times 10^7)/5$ $(= 1.26 \times 10^7 \text{ J kg}^{-1})$ — C1
change in energy $= (1.26 - 1.05) \times 10^7 \times 1.3$ — C1
$= 2.7 \times 10^6$ J — A1

or change in potential $= (1/5 - 1/6) \times (6.3 \times 10^7)$ — (C1)
change in energy $= (1/5 - 1/6) \times (6.3 \times 10^7) \times 1.3$ — (C1)
$= 2.7 \times 10^6$ J — (A1)

16.

(a) gravitational force provides/is the centripetal force — B1
$GMm/r^2 = mv^2/r$ — M1
$v = \sqrt{(GM/r)}$ — A0

allow gravitational field strength provides/is the centripetal acceleration — (B1)
$GM/r^2 = v^2/r$ — (M1)

(b) (i) kinetic energy increase/change = loss/change in (gravitational) potential energy — B1
$\tfrac{1}{2}mV_0^2 = GMm/x$ — C1
$V_0^2 = 2GM/x$
$V_0 = \sqrt{(2GM/x)}$ — A1

(max. 2 for use of r not x)

(ii) V_0 is (always) <u>greater than v</u> (for $x = r$) — M1
so stone could not enter into orbit — A1

(*expressions in **(a)** and **(b)(i)** must be dimensionally correct)*

17.

(a) $g = GM/R^2$ C1

$= (6.67 \times 10^{-11} \times 6.4 \times 10^{23})/(3.4 \times 10^6)^2 = 3.7\,\text{N kg}^{-1}$ A1

(b) $\Delta E_P = mg\Delta h$

because $\Delta h \ll R$ (or $1800\,\text{m} \ll 3.4 \times 10^6\,\text{m}$) g is constant B1

$\Delta E_P = 2.4 \times 3.7 \times 1800$ C1

$= 1.6 \times 10^4\,\text{J}$ A1

(*use of $g = 9.8\,\text{m s}^{-2}$ max. 1 for explanation*)

(c) gravitational potential <u>energy</u> = $(-)GMm/x$ C1

$v^2 = 2GM/x$ C1

$x = 4D = 4 \times 6.8 \times 10^6$ C1

$v^2 = (2 \times 6.67 \times 10^{-11} \times 6.4 \times 10^{23})/(4 \times 6.8 \times 10^6)$

$= 3.14 \times 10^6$

$v = 1.8 \times 10^3\,\text{m s}^{-1}$ A1

(*use of $3.5D$ giving $1.9 \times 10^3\,\text{m s}^{-1}$, allow max. 3*)

18.

(a) (gravitational) force proportional to product of masses and inversely proportional to square of separation M1

reference to *either* point masses *or* particles *or* 'size' much less than separation A1

(b) gravitational force provides/is the centripetal force B1

$GM_N m/r^2 = mr\omega^2$ (or mv^2/r) M1

$2\pi/T$ (*or* $v = 2\pi r/T$) leading to $GM_N = 4\pi^2 r^3/T^2$ A1

(c) $M_N/M_U = (3.55/5.83)^3 \times (13.5/5.9)^2$

x^3 factor correct C1

T^2 factor correct C1

ratio = 1.18 (*allow 1.2*) A1

alternative method: mass of Neptune = $1.019 \times 10^{26}\,\text{kg}$ (C1)

mass of Uranus = $8.621 \times 10^{25}\,\text{kg}$ (C1)

ratio = 1.18 (A1)

19.

(a) **(i)** **1.** $F = Gm_1m_2/x^2$
$= (6.67 \times 10^{-11} \times 2.50 \times 5.98 \times 10^{24})/(6.37 \times 10^6)^2$ M1
$= 24.6\,\text{N}$ (*accept 2 s.f. or more*) A1

2. $F = mx\omega^2$ **or** $F = mv^2/x$ **and** $v = \omega x$ *(accept x or r for distance)* C1
$= 2.50 \times 6.37 \times 10^6 \times (2\pi/24 \times 3600)^2$
$= 0.0842\,\text{N}$ (*accept 2 s.f. or more*) A1

(ii) reading $= 24.575 - 0.0842$ B1
$= 24.5\,\text{N}$ (*accept only 3 s.f.*) A1

(b) gravitational force provides the centripetal force M1
gravitational force is 'equal' to the centripetal force
(*accept* $Gm_1m_2/x^2 = mx\omega^2$ **or** $F_C = F_G$) M1
'weight'/sensation of weight/contact force/reaction force is difference between F_G and F_C which is zero A1

20.

(a) **(i)** gravitational force provides/is the centripetal force B1

$GMm_S/x^2 = m_Sv^2/x$ (allow x or r; allow m or m_S) M1

$E_K = \frac{1}{2}m_Sv^2$ and clear algebra leading to $E_K = GMm_S/2x$ A1

(ii) $E_P = -\,GMm_S/x$ (*sign essential*) B1

(iii) $E_T = E_K + E_P$
$= GMm_S/2x - GMm_S/x$ C1
$= -\,GMm_S/2x$ (*allow ECF from* ***(a)(ii)***) A1

(b) **(i)** decreases B1

(ii) decreases B1

(iii) decreases B1

(iv) increases B1

(*for answers in* ***(b)*** *allow ECF from* ***(a)(iii)***)

21.

(a) (gravitational) force proportional to product of masses and inversely proportional to square of separation — M1

either point masses *or* particles *or* 'size' ≪ separation — A1

(b) gravitational force provides the centripetal force — B1

either $GMm/x^2 = mx\omega^2$ *or* mv^2/x — M1

either $\omega = 2\pi/T$ *or* $v = 2\pi x/T$ and working to $GM = 4\pi^2x^3/T^2$ — A1

(c) *either* use of gradient of graph *or* line through origin so can use single point *or* line shown extrapolated to origin — B1

gradient = $(4.5 \times 10^{14})/0.35$

$6.67 \times 10^{-11} \times M = 4\pi^2 \times (4.5 \times 10^{14} \times 10^9)/(0.35 \times \{24 \times 3600\}^2)$

correct conversion for km^3 and power of 10 — C1

correct conversion for day^2 — C1

$M = 1.02 \times 10^{26}$ kg — A1

22.

(a) force proportional to product of the (two) masses and inversely proportional to the square of their separation — M1

either reference to point masses *or* separation << 'size' of masses — A1

(b) gravitational force provides/is the centripetal force — B1

$GMm/r^2 = mv^2/r$ *or* $GMm/r^2 = mr\omega^2$ and $v = r\omega$ and algebra leading to $v = (GM/r)^{1/2}$ — B1

(c) **(i)** **1.** $v_A/v_B = (r_B/r_A)^{1/2}$

$= (2.2 \times 10^{10}/1.3 \times 10^8)^{1/2}$ — C1

$= 13\ (13.0)$ — A1

2. $v = 2\pi r/T$ *or* $v \propto r/T$ *or* vT/r = constant — C1

$T_A/T_B = (r_A/r_B) \times (v_B/v_A)$

$= (1.3 \times 10^8/2.2 \times 10^{10}) \times (1/13)$ — C1

$= 4.5\ (4.54) \times 10^{-4}$ — A1

or

$T^2 = 4\pi^2 r^3 / GM$ *or* $T^2 \propto r^3$ *or* T^2 / r^3 = constant (C1)

$T_A / T_B = (r_A^3 / r_B^3)^{1/2}$

$= [(1.3 \times 10^8)^3 / (2.2 \times 10^{10})^3]^{1/2}$ (C1)

$= 4.5\ (4.54) \times 10^{-4}$ (A1)

(ii) $T = 2\pi / 1.7 \times 10^{-4}$

$= 3.70 \times 10^4$ s C1

$T_B = 3.70 \times 10^4 / 4.54 \times 10^{-4}$

$= 8.1 \times 10^7$ s A1

If identifies T_A as T_B then 0/2

23.

(a) (gravitational) potential at infinity defined as/is zero B1

(gravitational) force <u>attractive</u> so work got out/done as object moves from infinity (so potential is negative) B1

(b) **(i)** $\Delta E = m\Delta\phi$

$= 180 \times (14 - 10) \times 10^8$ C1

$= 7.2 \times 10^{10}$ J A1

increase B1

(ii) energy required = $180 \times (10 - 4.4) \times 10^8$

or

energy per unit mass = $(10 - 4.4) \times 10^8$ C1

$\frac{1}{2} \times 180 \times v^2 = 180 \times (10 - 4.4) \times 10^8$

or

$\frac{1}{2} \times v^2 = (10 - 4.4) \times 10^8$ C1

$v = 3.3 \times 10^4$ m s^{-1} A1

24.

(a) **(i)** gravitational force provides/is the centripetal force B1

same gravitational force (by Newton III) B1

(ii) $\omega = 2\pi / T$

$= 2\pi / (4.0 \times 365 \times 24 \times 3600)$ C1

$= 5.0\ (4.98) \times 10^{-8}$ rad s^{-1} A1

(b) (i) (centripetal force =) $M_A d\omega^2 = M_B(2.8 \times 10^8 - d)\omega^2$
or
$M_A d_A = M_B d_B$ C1

$M_A / M_B = 3.0 = (2.8 \times 10^8 - d) / d$ C1

$d = 7.0 \times 10^7$ km A1

(ii) $GM_A M_B / (2.8 \times 10^{11})^2 = M_A d\omega^2$ B1

$M_B = (2.8 \times 10^{11})^2 \times d\omega^2 / G$
$= (2.8 \times 10^{11})^2 \times (7.0 \times 10^{10}) \times (4.98 \times 10^{-8})^2 / (6.67 \times 10^{-11})$ C1

$= 2.0 \times 10^{29}$ kg A1

25.

(a) gravitational force provides/is the centripetal force B1

$GMm / r^2 = mv^2 / r$ *or* $GMm / r^2 = mr\omega^2$
and $v = 2\pi r / T$ *or* $\omega = 2\pi / T$ M1

with algebra to $T^2 = 4\pi^2 r^3 / GM$ A1

or

acceleration due to gravity is the centripetal acceleration (B1)

$GM / r^2 = v^2 / r$ *or* $GM / r^2 = r\omega^2$
and $v = 2\pi r / T$ *or* $\omega = 2\pi / T$ (M1)

with algebra to $T^2 = 4\pi^2 r^3 / GM$ (A1)

(b) (i) equatorial orbit/orbits (directly) above the equator B1

from west to east B1

(ii) $(24 \times 3600)^2 = 4\pi^2 r^3 / (6.67 \times 10^{-11} \times 6.0 \times 10^{24})$ C1

$r^3 = 7.57 \times 10^{22}$

$r = 4.2 \times 10^7$ m A1

(c) $(T/24)^2 = \{(2.64 \times 10^7)/(4.23 \times 10^7)\}^3$ B1
$= 0.243$

$T = 12$ hours A1

or

$k\ (= T^2/r^3) = 24^2/(4.23 \times 10^7)^3$ (B1)
$= 7.61 \times 10^{-21}$

$T^2\ (= kr^3) = 7.61 \times 10^{-21} \times (2.64 \times 10^7)^3$
$= 140$

$T = 12$ hours (A1)

26.

(a) force per unit mass B1

(b) **(i)** radius/diameter/size (of Proxima Centauri) $\ll$ /is <u>much</u> less than 4.0×10^{13} km/separation (of Sun and star)
or
(because) it is a <u>uniform</u> sphere B1

(ii) **1.** field strength $= GM/x^2$

$= (6.67 \times 10^{-11} \times 2.5 \times 10^{29})/(4.0 \times 10^{13} \times 10^3)^2$ C1

$= 1.0 \times 10^{-14}\ \text{N kg}^{-1}$ A1

2. force = field strength $\times$ mass

$= 1.0 \times 10^{-14} \times 2.0 \times 10^{30}$ C1

or

force $= GMm/x^2$

$= (6.67 \times 10^{-11} \times 2.5 \times 10^{29} \times 2.0 \times 10^{30})/(4.0 \times 10^{13} \times 10^3)^2$ (C1)

$= 2.0 \times 10^{16}$ N A1

(c) force (of 2×10^{16} N) would have little effect on (large) mass of Sun B1

would cause an acceleration of Sun of 1.0×10^{-14} m s^{-2}/very small/negligible acceleration B1

or

many stars all around the Sun (B1)
net effect of forces/fields is zero (B1)

27.

1(a)	gravitational force (of attraction between satellite and planet)	**B1**
	provides / is centripetal force (on satellite about the planet)	**B1**
1(b)	$M = (4/3) \times \pi R^3 \rho$	**B1**
	$\omega = 2\pi / T$ **or** $v = 2\pi nR / T$	**B1**
	$GM / (nR)^2 = nR\omega^2$ **or** v^2 / nR	**M1**
	substitution clear to give $\rho = 3\pi n^3 / GT^2$	**A1**
1(c)	$n = (3.84 \times 10^5) / (6.38 \times 10^3) = 60.19$ or 60.2	**C1**
	$\rho = 3\pi \times 60.19^3 / [(6.67 \times 10^{-11}) \times (27.3 \times 24 \times 3600)^2]$	**C1**
	$\rho = 5.54 \times 10^3$ kg m^{-3}	**A1**

28.

1(a)	force per unit mass	**B1**
1(b)(i)	$g = GM / r^2$ $= (6.67 \times 10^{-11} \times 1.0 \times 10^{13}) / (3.6 \times 10^3)^2$	**C1**
	$= 5.1 \times 10^{-5}$ N kg^{-1}	**A1**
1(b)(ii)	mass = (960 / 9.81) kg weight on comet = $(960 / 9.81) \times 5.1 \times 10^{-5}$	**C1**
	$= 5.0 \times 10^{-3}$ N	**A1**
1(c)	similarity: e.g. both attractive/pointed towards the comet e.g. same order of magnitude	**B1**
	difference: e.g. radial/non-radial e.g. same (over surface)/varies (over surface)	**B1**

29.

3(a)	force per unit mass	**B1**
3(b)	changes in height much less than radius of Earth	**M1**
	so (radial) field lines are almost parallel **or** $g = GM / R^2 \approx GM / (R + h)^2$	**A1**

3(c)	gravitational force provides/is centripetal force	**B1**
	$GMm/r^2 = mv^2/r$	**C1**
	$v = (2\pi \times 1.5 \times 10^{11})/(3600 \times 24 \times 365) = 2.99 \times 10^4$ (m s^{-1})	**C1**
	$6.67 \times 10^{-11}M = 1.5 \times 10^{11} \times (2.99 \times 10^4)^2$	**C1**
	$M = 2.0 \times 10^{30}$ kg	**A1**
	or	
	$GMm/r^2 = mr\omega^2$	**(C1)**
	$\omega = 2\pi/(3600 \times 24 \times 365) = 1.99 \times 10^{-7}$ (rad s^{-1})	**(C1)**
	$6.67 \times 10^{-11}M = (1.5 \times 10^{11})^3 \times (1.99 \times 10^{-7})^2$	**(C1)**
	$M = 2.0 \times 10^{30}$ kg	**(A1)**
	or	
	$T^2 = 4\pi^2r^3/GM$	**(C2)**
	$M = 4\pi^2 \times (1.5 \times 10^{11})^3 / (\{3600 \times 24 \times 365\}^2 \times 6.67 \times 10^{-11})$	**(C1)**
	$= 2.0 \times 10^{30}$ kg	**(A1)**

30.

1(a)	<u>force</u> proportional to <u>product</u> of masses and inversely proportional to square of separation	**B1**
	idea of <u>force</u> between <u>point</u> masses	**B1**
1(b)	mass of Jupiter (M) = $(4/3)\pi R^3\rho$	**B1**
	$\omega = 2\pi/T$ **or** $v = 2\pi nR/T$	**B1**
	$(m)\omega^2x = GM(m)/x^2$ **or** $(m)v^2/x = GM(m)/x^2$	**M1**
	substitution and correct algebra leading to $\rho T^2 = 3\pi n^3/G$	**A1**
1(c)(i)	$n = (4.32 \times 10^5)/(7.15 \times 10^4)$ **or** $n = 6.04$	**C1**
	$\rho \times (42.5 \times 3600)^2 = (3\pi \times 6.04^3)/(6.67 \times 10^{-11})$	**C1**
	$\rho = 1.33 \times 10^3$ kg m^{-3}	**A1**
1(c)(ii)	Jupiter likely to be a gas/liquid (at high pressure) [allow other sensible suggestions]	**B1**

31.

1(a)(i)	*either* direction of force on a (small test) mass *or* direction of acceleration of a (small test) mass	**B1**
1(a)(ii)	Any three from: • the lines are radial • near the surface the lines are (approximately) parallel • parallel lines so constant field strength • constant field strength hence constant acceleration of free fall	**B3**
1(b)(i)	$g = GM/R^2$ $g = (6.67 \times 10^{-11} \times 7.35 \times 10^{22}) / (1.74 \times 10^3 \times 10^3)^2$	**C1**
	$g =$ **1.62 N kg^{-1}**	**A1**
1(b)(ii)	*either* $x\omega^2 = GM/x^2$ <u>and</u> $\omega = 2\pi/T$ *or* $v^2/x = GM/x^2$ <u>and</u> $v = 2\pi r/T$	**C1**
	$(1.74 \times 10^6 + 320 \times 10^3)^3 \times 4\pi^2 / T^2 = (6.67 \times 10^{-11} \times 7.35 \times 10^{22})$	**C1**
	$T^2 = 7.04 \times 10^7$ $T =$ **8400 s** (8390)	**A1**

32.

1(a)(i)	direction of force on a (small test) <u>mass</u> **or** path in which a (small test) <u>mass</u> will move	**B1**
1(a)(ii)	(at surface,) lines (of force) are radial	**B1**
	Earth has large radius/height above surface is small so lines are (approximately) parallel	**B1**
	parallel lines → constant field strength	**B1**
1(b)	(change in) KE of rock = (change in) PE **or** $\frac{1}{2}mv^2 = GMm/R$	**C1**
	$(m)v^2 = (m)(2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22}) / (1.7 \times 10^3 \times 10^3)$	**C1**
	$v = 2.4 \times 10^3\,\mathrm{m\,s^{-1}}$	**A1**
	correct conclusion based on <u>comparison</u> of v with 2.8 km s^{-1}	**B1**
	or	
	(change in) KE of rock = (change in) PE	**(C1)**
	(at infinity) $E_P = (6.67 \times 10^{-11} \times 7.4 \times 10^{22} \times m) / (1.7 \times 10^3 \times 10^3)$ $= 2.9 \times 10^6\,m$	**(C1)**
	E_K of rock $= \frac{1}{2} \times m \times (2.8 \times 10^3)^2 = 3.9 \times 10^6\,m$	**(A1)**
	correct conclusion based on <u>comparison</u> of E_K and E_P values	**(B1)**
	or	

	(change in) KE of rock = (change in) PE **or** $\frac{1}{2}mv^2 = GMm/R$	**(C1)**
	$(m)\,(2800)^2 = (m)\,(2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22})/R$	**(C1)**
	$R = 1.3 \times 10^3$ km	**(A1)**
	correct conclusion based on <u>comparison</u> of R with 1.7×10^3 km	**(B1)**
	or	
	(change in) KE of rock = (change in) PE **or** $\frac{1}{2}mv^2 = GMm/R$	**(C1)**
	$(m)\,(2800)^2 = (m)\,(2 \times 6.67 \times 10^{-11} \times M)/(1.7 \times 10^6)$	**(C1)**
	$M = 1.0 \times 10^{23}$ kg	**(A1)**
	correct conclusion based on <u>comparison</u> of M with 7.4×10^{22} kg	**(B1)**

33.

1(a)(i)	work done per unit mass	**B1**
	work done moving mass from infinity (to the point)	**B1**
1(a)(ii)	(near Earth's surface change in) height $\ll$ radius **or** height <u>much</u> less than radius	**B1**
	potential inversely proportional to radius **and** radius approximately constant (so potential approximately constant)	**B1**
1(b)	initial kinetic energy = (–) potential energy (at surface) **or** $\frac{1}{2}mv^2 = GMm/r$	**B1**
	$v^2 = (2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22})/(0.5 \times 3.5 \times 10^6)$	**C1**
	$v = 2.4 \times 10^3\ \mathrm{m\,s^{-1}}$	**A1**

34.

1(a)(i)	force per unit mass	**B1**
1(a)(ii)	acceleration = F/m, field strength = F/m, so equal	**B1**
1(b)	smooth curve between R and $4R$ with negative gradient of decreasing magnitude	**B1**
	line passing through (R, 1.00g) and (2R, 0.25g)	**B1**
	line ending at (4R, 0.0625g)	**B1**
1(c)	$M = (4/3 \times \pi R^3)\rho$	**C1**
	$g = GM/(2R)^2$	**C1**
	$g = \frac{1}{3} \times 6.67 \times 10^{-11} \times \pi \times 3.4 \times 10^6 \times 4.0 \times 10^3$ $= 0.95\ \mathrm{m\,s^{-2}}$	**A1**

35.

1(a)(i)	work done per unit mass	**B1**
	idea of work done moving mass from infinity (to the point)	**B1**
1(a)(ii)	(gravitational) force is attractive	**B1**
	(gravitational) potential at infinity is zero	**B1**
	decrease in potential energy as masses approach *or* displacement and force in opposite directions	**B1**
1(b)(i)	*Either* $mv^2 / R = GMm / R^2$ *Or* $v = \sqrt{(GM / R)}$ $v^2 = (6.67 \times 10^{-11} \times 6.00 \times 10^{24}) / (7.30 \times 10^6)$	**C1**
	giving $v = 7.4 \times 10^3\ \mathrm{m\ s^{-1}}$	**A1**
1(b)(ii)	$V_P = -GMm / R$	**C1**
	$= -(6.67 \times 10^{-11} \times 6.00 \times 10^{24} \times 340) / (7.30 \times 10^6)$	**C1**
	$V_P = -1.9 \times 10^{10}\ \mathrm{J}$	**A1**
1(c)	$v^2 \propto 1 / r$, (r smaller) so v greater	**M1**
	and E_K greater	**A1**

36.

1(a)	(F =) GMm / x^2, where G is the (universal) gravitational constant	**B1**
1(b)(i)	$GMm / x^2 = mv^2 / x$ **or** $v^2 = GM / x$	**C1**
	$v^2 = (6.67 \times 10^{-11} \times 7.5 \times 10^{23}) / (3.4 \times 10^6 + 240 \times 10^3)$ so $v = 3.7 \times 10^3\ \mathrm{m\ s^{-1}}$	**A1**
1(b)(ii)	potential energy = $(-)GMm / x$	**C1**
	$E_A = (-)(6.67 \times 10^{-11} \times 7.5 \times 10^{23} \times 650) / (3.64 \times 10^6)$ **or** $E_B = (-)(6.67 \times 10^{-11} \times 7.5 \times 10^{23} \times 650) / (5.00 \times 10^7)$	**M1**
	correct substitution and subtraction $E_B - E_A$ shown, leading to $\Delta E_p = 8.3 \times 10^9\ \mathrm{J}$	**A1**
	or	
	$\phi = (-)GM / x$ **and** potential energy = $m\phi$	**(C1)**
	$\Delta\phi = (6.67 \times 10^{-11} \times 7.5 \times 10^{23}) \times [(1 / (3.64 \times 10^6)) - (1 / (5.00 \times 10^7))]$ $(= 1.27 \times 10^7\ \mathrm{J\ kg^{-1}})$	**(M1)**
	$\Delta E_p = 1.27 \times 10^7 \times 650$ $= 8.3 \times 10^9\ \mathrm{J}$	**(A1)**
1(c)	kinetic energy <u>or</u> potential energy decreases	**B1**
	kinetic energy <u>and</u> potential energy decrease so total energy decreases	**B1**

37.

1(a)	(F =) GMm / x^2, where G is the (universal) gravitational constant	**B1**
1(b)(i)	angle = $(1.2 \times 10^{-3}) / (8.0 \times 10^{-2}) = 1.5 \times 10^{-2}$ (rad)	**B1**
1(b)(ii)	torque $= 1.5 \times 10^{-2} \times 9.3 \times 10^{-10}$ $= 1.4 \times 10^{-11}$ N m	**A1**
1(c)(i)	force $\times 8.0 \times 10^{-2} = 1.4 \times 10^{-11}$	**C1**
	$(G \times 1.3 \times 7.5 \times 10^{-3} \times 8.0 \times 10^{-2}) / (6.0 \times 10^{-2})^2 = 1.4 \times 10^{-11}$	**C1**
	$G = 6.4 \times 10^{-11}$ N m^2 kg^{-2}	**A1**
1(c)(ii)	Any one from: • law applies only to point masses/spheres are not point masses • radii of spheres not small compared with separation • spheres may not be uniform • the masses are not isolated • force between L and rod • spheres may be charged/may be electrostatic force (between spheres)	**B1**

38.

1(a)	force proportional to product of masses and inversely proportional to square of separation	**B1**
	idea of (gravitational) force between point masses	**B1**
1(b)(i)	above the equator	**B1**
	from west to east	**B1**
1(b)(ii)	gravitational force provides/is the centripetal force	**B1**
	$GM / r^2 = r(2\pi / T)^2$	**C1**
	$(6.67 \times 10^{-11} \times M) = \{(4.23 \times 10^7)^3 \times 4\pi^2\} / (24 \times 3600)^2$	**C1**
	$M = 6.0 \times 10^{24}$ kg	**A1**

39.

1(a)	force proportional to product of masses and inversely proportional to square of separation	**B1**
	idea of (gravitational) force between point masses	**B1**
1(b)	gravitational force provides/is the centripetal force	**B1**
	$GM / R^2 = R\omega^2$ **or** $GM / R^2 = v^2 / R$	**M1**
	$\omega = 2\pi / T$ **or** $v = 2\pi R / T$	**M1**
	algebra leading to $R^3/T^2 = GM / 4\pi^2$	**A1**
1(c)	$(6.67 \times 10^{-11} \times M) / 4\pi^2 = (4.38 \times 10^6)^3 / (2.44 \times 3600)^2$	**C1**
	$M = 6.45 \times 10^{23}$ kg	**A1**

40.

1(a)	work done per unit mass	**B1**
	work done moving mass from infinity (to the point)	**B1**
1(b)(i)	gravitational force provides centripetal force	**C1**
	$mv^2 / r = GMm / r^2$ and $v = 2\pi r / T$ OR $mr\omega^2 = GMm / r^2$ and $\omega = 2\pi / T$ OR $r^3 = GMT^2 / 4\pi^2$	**C1**
	$r^3 = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (13.7 \times 24 \times 3600)^2 / 4\pi^2$ so $r = 2.4 \times 10^8$ m	**A1**
1(b)(ii)	($E_P = -$) GMm / r work done = $GMm / r_1 - GMm / r_2$	**C1**
	$= 6.67 \times 10^{-11} \times 360 \times 6.0 \times 10^{24}$ $(1/6.4 \times 10^6 - 1 / 2.4 \times 10^8)$	**C1**
	= $\mathbf{2.2 \times 10^{10}}$ J	**A1**
1(b)(iii)	$g = GM / r^2$	**C1**
	ratio $= r_{TESS}^2 / r_{earth}^2$ $= (2.4 \times 10^8 / 6.4 \times 10^6)^2$ = **1400**	**A1**

41.

1(a)	work done per unit mass	**B1**
	(work done to) move mass from infinity (to the point)	**B1**
1(b)	curve from r to $4r$, with gradient of decreasing magnitude and starting at (r, $\pm\phi$)	**B1**
	line passing through ($2r$, $\pm 0.5\phi$) and ($4r$, $\pm 0.25\phi$)	**B1**
	line showing potential is negative throughout	**B1**
1(c)(i)	gravitational potential energy = (–) GMm / R	**C1**
	change $= (6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 3.4 \times 10^3) / (6.4 \times 10^6) \times [1/3 - 1/4]$	**C1**
	$= 1.8 \times 10^{10}$ J	**A1**
1(c)(ii)	rock loses potential energy	**B1**
	(so) kinetic energy increases so speed increases	**B1**
	or	
	force is attractive	**(B1)**
	moves towards planet so speeds up	**(B1)**

42.

1(a)(i)	region (of space)	**B1**
	where a particle experiences a force	**B1**
1(a)(ii)	force per unit mass	**B1**
1(b)	$g = GM / R^2$	**C1**
	$= (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / (3.39 \times 10^6)^2$	**C1**
	$= 3.73$ N kg^{-1}	**A1**

1(c)	$0.99 \times 3.73 = (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / r^2$	C1
	$r = 3.41 \times 10^6$ (m)	C1
	height = $(r - R)$ $= 2 \times 10^4$ m	A1
	or	
	$0.99 \times 3.73 = (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / (R + h)^2$ $(R + h)^2 = 1.1596 \times 10^{13}$	(C1)
	$R + h = 3.41 \times 10^6$ (m)	(C1)
	$h = 2 \times 10^4$ m	(A1)
	or	
	$0.99 = (3.39 \times 10^6)^2 / r^2$	(C1)
	$r = 3.41 \times 10^6$ (m)	(C1)
	height = 2×10^4 m	(A1)

43.

1(a)	work done per unit mass	B1
	(work done) moving mass from infinity (to the point)	B1
1(b)(i)	gravitational potential energy = $(-)GMm / r$	C1
	$\Delta E_P = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 2.4 \times 10^3 \times [(6.4 \times 10^6)^{-1} - (1.2 \times 10^7)^{-1}]$	C1
	or	
	$\Delta\phi = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times [(6.4 \times 10^6)^{-1} - (1.2 \times 10^7)^{-1}]$	(C1)
	$\Delta E_P = m\Delta\phi$	(C1)
	$\Delta E_P = 7.0 \times 10^{10}$ J	A1
1(b)(ii)	$GMm / r^2 = mv^2 / r$	C1
	$v^2 = GM / r$ $= (6.67 \times 10^{-11} \times 6.0 \times 10^{24}) / (1.2 \times 10^7)$	C1
	$v = 5800$ m s^{-1}	A1
1(c)	any one point from: • smaller gain in energy required if orbit is west to east • smaller change in velocity if orbit is west to east • smaller gain in energy if orbit is in same direction as Earth's rotation • smaller change in velocity if orbit is in same direction as Earth's rotation • satellite already moving west to east at launch • Earth's rotation is from west to east	B1

44.

1(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(b)	correct read offs from the graph with correct power of ten for R^3	C1
	$M = \frac{4\times\pi^2\times1.2\times10^{34}}{6.67\times10^{-11}\times2.4\times(365\times24\times3600)^2}$	C1
	$=3.0\times10^{30}$ kg	A1
1(c)(i)	potential energy is zero at infinity	B1
	(gravitational) forces are attractive	B1
	work must be done on the rock to move it to infinity	B1
1(c)(ii)	$\frac{GMm}{r^2} = \frac{mv^2}{r}$ *OR* $v^2 = \frac{GM}{r}$ *OR* $v = \sqrt{\frac{GM}{r}}$	M1
	use of ½ mv^2 (e.g. multiplication by ½ m) leading to $\frac{GMm}{2r}$	A1
1(c)(iii)	Ep = ϕm <u>and</u> $\phi = \frac{-GM}{r}$ or $E_p = \frac{-GMm}{r}$ Total energy = $E_k + E_p$	C1
	Total energy $= \frac{GMm}{2r} + \frac{-GMm}{r} = \frac{-GMm}{2r}$	A1

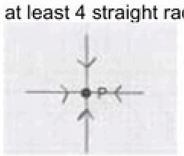
45.

1(a)	force per unit mass	B1
1(b)	$GMm / r^2 = mr\omega^2$ **and** $\omega = 2\pi/T$ **or** $GMm / r^2 = mv^2 / r$ **and** $v = 2\pi r / T$	C1
	$6.67 \times 10^{-11} \times 6.0 \times 10^{24} = r^3 \times [2\pi / (94 \times 60)]^2$	C1
	$r = 6.9 \times 10^6$ m	A1
1(c)(i)	$r^3\omega^2$ = constant **or** r^3 / T^2 = constant	C1
	$r^3 / (6.9 \times 10^6)^3 = (150 / 94)^2$ so $r = 9.4 \times 10^6$ m	A1
	or	
	$GMT^2/4\pi^2 = r^3$ **and** clear that M is 6.0×10^{24}	(C1)
	$6.67 \times 10^{-11} \times 6.0 \times 10^{24} = r^3 \times [2\pi / (150 \times 60)]^2$ so $r = 9.4 \times 10^6$ m	(A1)
1(c)(ii)	separation increases so (potential energy) increases **or** movement is against gravitational force so (potential energy) increases	B1
1(c)(iii)	potential energy = $(-)GMm / r$	C1
	$\Delta E_P = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 1200 \times [(6.9 \times 10^6)^{-1} - (9.4 \times 10^6)^{-1}]$	C1
	$= 1.9 \times 10^{10}$ J	A1

46.

1(a)	(gravitational) force per unit mass	B1
1(b)(i)	$g = GM / r^2$	C1
	$= (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / (3.39 \times 10^6)^2$ $= 3.73\ \text{N kg}^{-1}$	A1
1(b)(ii)	$a = r\omega^2$ **and** $\omega = 2\pi / T$ **or** $a = v^2 / r$ **and** $v = 2\pi r / T$	C1
	$a = 3.39 \times 10^6 \times (2\pi / (24.6 \times 3600))^2$ $= 0.0171\ \text{m s}^{-2}$	A1
1(b)(iii)	force per unit mass $= 3.73 - 0.0171$ $= 3.71\ \text{N kg}^{-1}$	A1

47.

1(a)	at least 4 straight radial lines to P  	B1
	all arrows pointing along the lines towards P	B1
1(b)	**Any 2 from:** gravitational force provides the centripetal force (centripetal or gravitational) force has constant magnitude (centripetal or gravitational) force is perpendicular to velocity (of moon) / direction of motion (of moon)	B2
1(c)(i)	$\frac{GMm}{r^2} = mr\omega^2$	M1
	$M = \frac{r^3\omega^2}{G}$ and gradient $= r^3\omega^2$ hence $M = \frac{\text{gradient}}{G}$ or $r^3 = GM \times 1/\omega^2$ so gradient $= GM$ hence $M = \frac{\text{gradient}}{G}$	A1
1(c)(ii)	$M = 4.1 \times 10^{23} / (6.0 \times 10^7 \times 6.67 \times 10^{-11}) = \mathbf{1.0 \times 10^{26}}$ kg	B1
1(c)(iii)	$\frac{GMm}{r^2} = \frac{mv^2}{r}$ $\frac{GM}{r} = v^2$	C1
	$v^2 = \frac{6.67 \times 10^{-11} \times 1.0 \times 10^{26}}{1.2 \times 10^8}$ $v^2 = 5.6 \times 10^7\ \text{m s}^{-1}$	C1
	$v = 7500\ \text{ms}^{-1}$	A1

48.

1(a)(i)	(gravitational) force is (directly) proportional to product of masses	**B1**
	force (between point masses) is inversely proportional to the square of their separation	**B1**
1(a)(ii)	$g = F / m$	**C1**
	$F = GMm / r^2$ and so $g = [GMm / r^2] / m = GM / r^2$	**A1**
1(b)(i)	$g = (6.67 \times 10^{-11} \times 7.35 \times 10^{22}) / (1.74 \times 10^{6})^2 = 1.62\ \text{N kg}^{-1}$	**A1**
1(b)(ii)	fields (due to Earth and the Moon) have equal magnitudes	**B1**
	fields (due to Earth and the Moon) are in opposite directions	**B1**
1(b)(iii)	distance of X from Earth = $(3.84 \times 10^{8} - x)$	**C1**
	$(G \times)\ 7.35 \times 10^{22} / x^2 = (G \times)\ 5.98 \times 10^{24} / (3.84 \times 10^{8} - x)^2$	**C1**
	$x = 3.8 \times 10^{7}$ m	**A1**

49.

1(a)(i)	work (done) per unit mass	**B1**
	work (done on mass) in moving mass from infinity (to the point)	**B1**
1(a)(ii)	$E_P = \phi m$ $E_P = (-\ GM / r) \times m = -\ GMm / r$ **or** $\phi = -\ GM / r$ **and_** $E_P = \phi m = -\ GMm / r$	**B1**
1(b)(i)	$\Delta E_P = 6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 2.20 \times 10^{14} \times [1 / (6.38 \times 10^{10}) - 1 / (8.44 \times 10^{11})]$	**C1**
	$= 4.23 \times 10^{23}$ J	**A1**
1(b)(ii)	(gravitational) force is attractive so decrease **or** (gravitational) force does work so decrease	**B1**
1(b)(iii)	$\Delta E_P = ½m(v_2^2 - v_1^2)$	**C1**
	$4.23 \times 10^{23} = ½ \times 2.20 \times 10^{14} \times (v^2 - 34\,100^2)$	**C1**
	$v\ (= 70\,800\ \text{m s}^{-1}) = 70.8\ \text{km s}^{-1}$	**A1**
1(c)	both PE and KE equations include m, so path is unchanged	**B1**

50.

1(a)	$F = (Gm_1m_2) / r^2$	**M1**
	where G is the gravitational constant	**A1**
1(b)	gravitational force provides the centripetal force	**B1**
	$mR\omega^2 = GMm / R^2$ **and** $\omega = 2\pi / T$ **or** $mv^2 / R = GMm / R^2$ **and** $v = 2\pi R / T$ **or** $4\pi^2 mR / T^2 = GMm / R^2$	**M1**
	correct completion of algebra to get $T^2 = (4\pi^2 / GM)\ R^3$, with identification of $(4\pi^2 / GM)$ as k	**A1**
1(c)(i)	$(24 \times 3600)^2 = (4\pi^2 \times R^3) / (6.67 \times 10^{-11} \times 6.0 \times 10^{24})$	**C1**
	$R = 4.2 \times 10^{7}$ m	**A1**
1(c)(ii)	(orbit) must be above the Equator	**B1**
	(direction) must be from west to east	**B1**

51.

1(a)	force per unit mass	B1
1(b)(i)	lines drawn are radial from the surface	B1
	arrows show pointing towards planet	B1
1(b)(ii)	field lines show force (on satellite) is towards centre of planet **or** velocity of satellite is perpendicular to field lines	B1
	(gravitational) force perpendicular to velocity causes centripetal <u>acceleration</u>	B1
1(c)(i)	T = 24 hours	C1
	$a = r\omega^2$ **and** $\omega = 2\pi / T$ **or** $a = v^2 / r$ **and** $v = 2\pi r / T$ **or** $a = 4\pi^2 r / T^2$	C1
	$a = (4\pi^2 \times 6.4 \times 10^6) / (24 \times 60 \times 60)^2$ $= 0.034\ \text{m s}^{-2}$	A1
1(c)(ii)	identification of the two forces acting on the object as gravitational force and (normal) contact force	M1
	gravitational force and normal contact force are in opposite directions, and their resultant causes the (centripetal) acceleration	A1

52.

1(a)	work done per unit mass	B1
	work (done on mass) moving mass from infinity (to the point)	B1
1(b)(i)	$-3.55 \times 10^7\ \text{J kg}^{-1}$	B1
1(b)(ii)	$\phi = -\dfrac{GM}{r}$ $M = -\dfrac{-3.55\times10^7 \times 4800000}{6.67\times10^{-11}}$ $= 2.55 \times 10^{24}$ kg	B1
1(b)(iii)	$g = \dfrac{GM}{r^2}$ **or** $g = -\dfrac{\phi}{r}$	C1
	$= \dfrac{6.67\times10^{-11}\times2.55\times10^{24}}{4800000^2}$ **or** $= \dfrac{3.55\times10^7}{4800000}$ $= 7.4\ \text{N kg}^{-1}$	A1
1(b)(iv)	r in range 2.60×10^7 to 2.65×10^7 m	C1
	$\dfrac{mv^2}{r} = \dfrac{GMm}{r^2}$ **and** $v = \dfrac{2\pi r}{T}$ **or** $mr\omega^2 = \dfrac{GMm}{r^2}$ **and** $\omega = \dfrac{2\pi}{T}$	C1
	$T^2 = \dfrac{4\pi^2 r^3}{GM} = \dfrac{4\pi^2 \times (2.65\times10^7)^3}{6.67\times10^{-11}\times2.55\times10^{24}} = 4.20 \times 10^9$	C1
	$T = 64\,800$ s $= 18$ hours	A1
1(c)	similarity – any one point from • inversely proportional to distance (from point) • points of equal potential lie on concentric spheres • zero at infinite distance	B1
	difference – any one point from • gravitational potential is (always) negative • electric potential can be positive or negative	B1

53.

1(a)(i)	force per unit mass	B1
1(a)(ii)	force per unit positive charge	B1
1(a)(iii)	similarity: • inversely proportional to distance (from point) • points of equal potential lie on concentric spheres • zero at infinite distance *Any point, 1 mark*	B1
	difference: • gravitational potential is (always) negative • electric potential can be positive or negative *Any point, 1 mark*	B1
1(b)(i)	$g = GM / r^2$	M1
	$E = Q / 4\pi\varepsilon_0 r^2$	M1
	algebra showing the elimination of r leading to $M / Q = (1 / 4\pi G\varepsilon_0)\ (g / E)$	A1
1(b)(ii)	$\alpha = 1 / (4\pi \times 6.67 \times 10^{-11} \times 8.85 \times 10^{-12}) = 1.35 \times 10^{20}$ ($kg^2\ C^{-2}$) **or** $\alpha = (8.99 \times 10^9) / (6.67 \times 10^{-11}) = 1.35 \times 10^{20}$ ($kg^2\ C^{-2}$)	A1
1(c)(i)	$E = \alpha g Q / M$ $= (1.35 \times 10^{20} \times 9.81 \times 4.80 \times 10^5) / (5.98 \times 10^{24})$	C1
	$= 106\ N\ C^{-1}$ **or** $106\ V\ m^{-1}$	A1
1(c)(ii)	same (direction)	B1

54.

1(a)	(gravitational) force is (directly) proportional to product of masses	B1
	force (between point masses) is inversely proportional to the square of their separation	B1
1(b)	$GMm / R^2 = mR\omega^2$	M1
	$\omega = 2\pi / T$ <u>and</u> algebra leading to $4\pi^2 R^3 = GMT^2$	A1
	or	
	$GMm / R^2 = mv^2 / R$	(M1)
	$v = 2\pi R / T$ <u>and</u> algebra leading to $4\pi^2 R^3 = GMT^2$	(A1)
1(c)	$4\pi^2 \times R^3 = 6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times (24 \times 60 \times 60)^2$ ($R = 4.22 \times 10^7$ m)	C1
	$h = R - (6.37 \times 10^6)$	C1
	$h = (4.22 \times 10^7) - (6.37 \times 10^6)$ $= 3.6 \times 10^7$ m	A1
1(d)(i)	$\omega = 2\pi / T$	C1
	$= 2\pi / (24 \times 60 \times 60)$ $= 7.3 \times 10^{-5}\ rad\ s^{-1}$	A1
1(d)(ii)	orbit is from east to west	B1
	orbit is not equatorial / orbit is polar	B1

55.

1(a)(i)	direction of the force acting on a (test) mass placed at the point	**B1**
1(a)(ii)	change in height negligible compared with radius (of Earth)	**B1**
	(so) field lines are (effectively) parallel	**B1**
1(b)(i)	$Y = GM / R^2$	**M1**
	G is the gravitational constant	**A1**
1(b)(ii)	gravitational force is (always) attractive **or** gravitational force (always) acts towards the centre of the sphere	**B1**
	force is in opposite direction to displacement **or** at a point to the right of the centre, force acts to the left **or** at a point to the left of the centre, force acts to the right	**B1**
1(b)(iii)	sketch: smooth curve with decreasing positive gradient, starting at (R, $-Y$) and reaching $3R$ with g still negative **or** smooth curve with increasing positive gradient, ending at ($-R$, Y) and reaching $-3R$ with g still positive	**B1**
	both of the above curves, in correct quadrants	**B1**
	curve passing through ($\pm 2R$, $\pm 0.25Y$) and ($\pm 3R$, $\pm 0.11Y$)	**B1**